

Introduction to Brain & Cognition

Comprehensive Lecture Notes

Lectures: March 4, 6, 10, 13, 17, 24, 27 & April 7, 10, 14, 17, 21, 24 — 2026

IIIT Hyderabad

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Lecture 1: The Science of the Mind (March 4)

1.1 Course Overview

This course provides a broad knowledge and appreciation of **cognitive psychology**—the scientific study of how the mind acquires, stores, retrieves, and uses knowledge to influence behaviour. The course draws from two textbooks:

1. *Cognition: Exploring the Science of the Mind* (7th edition)
2. *Thinking 101* (based on a popular Yale course)

Cognition

Cognition (Latin: *cognoscere* — “to think, to know”) refers to the mental processes by which knowledge is acquired, stored, retrieved, and used. It encompasses perception, attention, memory, language, reasoning, problem-solving, and decision-making.

1.2 Psychology and its Branches

Psychology is the study of the *psyche* (mind). Major branches include:

- **Experimental Psychology** — uses experimental methods to understand mind and behaviour
- **Social Psychology** — how groups behave together; group mind and group behaviour
- **Physiological Psychology** — what brain areas are important for certain behaviours; the material basis of behaviour (neurotransmitters, brain regions)
- **Cognitive Psychology** — how knowledge influences behaviour; how knowledge is acquired, stored, and retrieved

1.3 Historical Epochs of Psychology

1.3.1 Epoch 1: Introspection (1800s)

Introspection

Introspection is the method of studying the mind by looking inward—observing one’s own thoughts, feelings, and mental processes, and trying to identify patterns. Pioneered by **William James** (two volumes based on introspection and observation).

Introspection borders on philosophy and psychology. It involves reflecting on one’s own thought processes: “How do I make decisions? How do I perceive objects?”

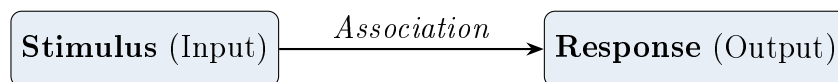
Limitations of Introspection

1. **Inconsistency** — the same question can yield different answers on different occasions
2. **Individual variability** — different people introspect and arrive at different conclusions
3. **Bias** — constrained by one's own experience and knowledge
4. **Cannot access unconscious processes** — introspection only works when you are awake and conscious
5. **Non-verifiable** — “you are the observer, the witness, and the judge”
6. Violates scientific requirements of *repeatability* and *verifiability*

1.3.2 Epoch 2: Behaviourism (~1900s–1950s)

Behaviourism

Behaviourism is a school of psychology that focuses exclusively on *observable behaviour*, dismissing the concept of a non-physical mind. All cognition is reduced to **stimulus–response (S–R) associations**. Prominent figure: **John B. Watson**.



Key principles of behaviourism:

- Behaviour types: **Avoidance behaviour** (retract from stimulus) and **Approach behaviour** (move toward stimulus)
- Stimuli can be from any input modality: visual, auditory, tactile, olfactory
- Repeated stimulus–response pairing strengthens the association (forming **habits**)
- Cognition is split into: (a) *Wired/genetic* (innate reflexes) and (b) *Learnt* (acquired S–R associations)
- “There is nothing called mind”—only reflexes and associations

Built-in reflexes (genetic):

- Eye-blink reflex (puff of air → blink)
- Withdrawal reflex (hand touches hot surface → retract)
- Crying when hungry (in infants)

Limitations of Behaviourism

1. **Stimulus generalisation problem** — “Pass the salt” vs. “Salt, please!” vs. “Could you pass the salt?” yield the same response, yet the stimuli are different. Behaviourism cannot explain how internal meaning mediates equivalent responses.
2. **Novel stimuli** — humans respond correctly to requests they have *never* heard before. Where does this generalisation come from if not from an internal “mind”?
3. **Cannot explain language acquisition** — children acquire language with very few examples, too fast for pure S–R learning.
4. **Cannot explain unconscious processing** or internal representations.

1.3.3 Epoch 3: The Cognitive Revolution (~1950s–60s)

Cognitive Revolution

The **Cognitive Revolution** was the paradigm shift from behaviourism to **cognitivism**, reintroducing the concept of the **mind** as an internal entity that mediates behaviour. The mind is no longer a taboo word.

Key contributors:

Edward Tolman — Cognitive Maps. Tolman studied rat navigation in mazes. Rats could generalise to *new paths* they had never visited, finding food efficiently. This implied an internal **cognitive map**—a stored representation useful in novel scenarios, not explainable by pure S–R.

George Miller — Chunking and Working Memory.

Miller’s Law: 7 ± 2

Working memory (the temporary buffer) has a limited capacity of approximately 7 ± 2 **chunks**. We overcome this limit by **chunking**—compressing information into meaningful units.

Example: A 10-digit phone number exceeds capacity. We break it into chunks: area code (chunk 1) + first group (chunk 2) + second group (chunk 3) = 3 chunks ≤ 7 .

Working Memory vs. Long-Term Memory

Working memory is a temporary scratch-pad with limited capacity (7 ± 2 chunks). Used for tasks like mental arithmetic.

Long-term memory can store vast amounts of information (e.g., entire poems) but requires repeated rehearsal and rote learning.

Noam Chomsky — Universal Grammar.

- Published a seminal book attacking behaviourism
- Argued that S–R associations *cannot* explain how children acquire language so rapidly from so few examples
- Proposed **Universal Grammar (UG)**: every brain is born with an innate, general-purpose grammar machine
- Exposure to a particular language (Telugu, Hindi, German, etc.) *specialises* this UG
- Also contributed the hierarchy of formal languages: context-free, context-sensitive grammars, etc.

1.4 Schemas and Scripts

Schema / Script

A **schema** (or **script**) is a mental framework—a compressed, structured representation of typical experiences—that helps us organise knowledge, generate expectations, and navigate the world in real time.

Example — Restaurant schema:

1. Enter restaurant → waiter asks how many people
2. Taken to a table → given a menu
3. Order food → waiter writes it down and reads it back
4. Food arrives → eat → pay bill

Example — Piggy bank schema:

“She shook it. It made no sound. She went to look for her mother.”

To interpret this, we need the schema: children receive money → store in piggy bank → shaking reveals contents by sound → no sound = no money → go to mother for money. Without this schema, the sentences are incomprehensible.

Counterfactual Reasoning

Counterfactual reasoning: “If X were the case, what would the consequence be?” This higher-order reasoning is largely unique to humans and is not well explained by S–R models. Animals (except possibly great apes) show limited counterfactual reasoning.

1.5 Real-Time Constraint on Cognition

Operating in the real world imposes severe **time constraints** on cognitive processing:

- A cricket batsman has only $\sim 15\text{--}50$ ms to decide how to play a delivery (full-toss, swing, etc.)
- Schemas and compressed representations allow rapid decision-making under time pressure

1.6 Research Methods in Cognitive Psychology

Cognitive psychology uses multiple methods:

1. **Behavioural experiments** — controlled lab tasks measuring accuracy, reaction time
2. **Eye tracking** — tracks position and movement of the eye; pupil dilation indicates cognitive load
3. **Physiological measurements** — galvanic skin conductance (GSR), heart rate variability (HRV)
4. **Neuroimaging** — EEG, fMRI, MEG, PET (detailed in later lectures)
5. **Computational modelling** — mathematical models of cognitive processes

Pupil Dilation and Cognitive Load

People engaged in *complex* tasks (e.g., 5-digit $\times$ 4-digit multiplication) show **pupil dilation**—an internal physiological reaction reflecting increased attention and cognitive effort. Simple tasks show no such dilation.

Practice & Review

Q1. [Multiple Choice] Which best defines *cognition*?

- (A) The study of brain physiology (B) How knowledge is acquired, stored, retrieved, and used to guide behaviour
(C) Observable stimulus–response behaviour only (D) Signal processing in the peripheral nervous system

Answer: B. Cognition (Latin *cognoscere*: “to know”) encompasses perception, attention, memory, language, and reasoning. It is the study of how knowledge is acquired, stored, retrieved, and used.

Q2. [Short Answer] What is introspection and what are four of its key limitations?

Answer: Introspection is the method of studying the mind by observing one’s own thoughts, feelings, and mental processes. Limitations: (1) **Inconsistency**—the same question yields different answers on different occasions; (2) **Individual variability**—different people arrive at different conclusions; (3) **Bias**—constrained

by personal experience and knowledge; (4) **No access to the unconscious**—introspection only operates when conscious; (5) **Non-verifiable**—the introspecting person is simultaneously observer and judge, violating scientific repeatability.

- Q3. [Essay]** Explain behaviourism, its core claim, and the three problems identified during the Cognitive Revolution. Reference Tolman, Miller, and Chomsky.

Answer: Behaviourism (Watson): all cognition reduces to stimulus–response (S–R) associations formed by repeated pairing; “mind” is inadmissible as a scientific concept. Key failures: (1) **Stimulus generalisation**—“Pass the salt,” “Salt, please,” and “Could you pass the salt?” all elicit the same response even though the stimuli differ; S–R theory cannot explain the internal *equivalence mapping*. (2) **Novel stimuli**—humans respond appropriately to requests never heard before. (3) **Language acquisition**—children acquire language too rapidly and from too few examples for pure S–R learning. Cognitive Revolution responses: **Tolman** demonstrated rats use internal *cognitive maps* to navigate new maze paths, implying stored internal representations; **Miller** established that working memory has a capacity of 7 ± 2 chunks (chunking = cognitive compression); **Chomsky** proposed Universal Grammar—an innate, general-purpose grammar machine specialised by early language exposure.

- Q4. [Multiple Choice]** Who proposed the theory of Universal Grammar?

(A) William James (B) Noam Chomsky (C) Edward Tolman (D) B.F. Skinner

Answer: **B.** Chomsky proposed that every brain is born with a universal grammar machine. Early exposure to a specific language (e.g., Telugu, Hindi, German) specialises it for that language’s rules and sounds.

- Q5. [Short Answer]** State Miller’s Law (7 ± 2) and explain how chunking allows a 10-digit phone number to be stored in working memory.

Answer: Working memory holds 7 ± 2 *chunks* of information. A raw 10-digit number (e.g., 9440055123) exceeds this limit. Chunking compresses it: area code **944** + group **0055** + group **123** = 3 chunks—well within the 7 ± 2 capacity. Chunking is a cognitive compression strategy that allows the limited buffer to store more information.

- Q6. [Short Answer]** What is a schema? Use the “piggy bank” example to explain why schemas are *necessary* for comprehending: “*She shook it. It made no sound. She went to look for her mother.*”

Answer: A schema (or script) is a compressed mental framework of typical experiences used to organise knowledge and generate expectations. The piggy bank schema: children receive money as gifts → store coins in a piggy bank → shaking reveals contents by sound → silence means empty → seek money from a parent. Without this schema the three sentences are individually comprehensible but their causal connection—and the inference that the child needs money—is impossible to derive.

Q7. [Multiple Choice] Counterfactual reasoning involves:

- (A) Recalling a specific past episode in detail
- (B) Imagining a hypothetical state of affairs and reasoning about its consequences
- (C) Forming a classical stimulus–response association
- (D) Recognising a familiar face in a crowd

Answer: B. Counterfactual reasoning (“if X were the case, what would happen?”) is a higher-order cognitive skill largely unique to humans; non-human animals show little evidence of it. It cannot be explained by S–R theory because there is no physical stimulus to respond to.

Q8. [Short Answer] Name four research methods used in cognitive psychology and state what each specifically measures.

Answer: (1) **Behavioural experiments**—accuracy and reaction time (RT) in controlled lab tasks; (2) **Eye tracking**—gaze position, fixation duration, and pupil dilation (a physiological proxy for cognitive load—pupil dilates for complex tasks); (3) **Physiological recording**—galvanic skin conductance (GSR) for arousal, heart rate variability (HRV) for autonomic state; (4) **Neuroimaging**—brain activity patterns via EEG, fMRI, MEG, or PET.

Lecture 2: Neural Basis for Cognition — Methods (March 6)

2.1 Recap: Three Epochs and the Fourth

Epoch	Approach	Era	Key Idea
1	Introspection	1700s–1800s	Look inward; subjective
2	Behaviourism	Early 1900s–1950s	S–R associations; objective
3	Cognitive Revolution	1950s–60s	Mind as mediator; internal representations
4	Consciousness	Emerging	Subjective experience; qualia

2.2 Attention

Attention

Attention is the cognitive ability to selectively focus on specific stimuli while inhibiting others. It enables prioritisation of relevant information.

The Cocktail Party Problem

In a noisy party with many simultaneous conversations, you can **focus on one speaker** and inhibit all others. This is **focused/selective attention**—a fundamental cognitive function.

Cognitive functions studied: perception, attention, inference, judgment, emotional feelings (happy, sad, angry), language, pattern recognition.

2.3 Consciousness and Qualia

Consciousness

Consciousness is the state of being aware of one's surroundings, thoughts, and emotions. It involves:

1. Being alive, awake, and able to respond to stimuli
2. The ability to **experience and reflect** on those experiences
3. Using reflection to guide future actions

Self-consciousness (the ability to reflect on one's own experience) is argued to be uniquely human.

Qualia

Qualia (singular: *quale*) refer to the *subjective, personal* quality of conscious experiences. E.g., the “crunchiness” and “sweetness” you experience when biting an apple is *your* personal quale—it cannot be directly shared or observed by others.

Consciousness as the 4th Epoch

Current science does not have a complete framework for consciousness. Like “mind” was taboo in the behaviourist epoch, consciousness is still not fully integrated into cognitive science. Perhaps 50–100 years from now, we will look back and wonder how we ignored something so fundamental.

Unconscious processes: Many functions happen unconsciously—walking, counting steps, breathing. We are not conscious of most low-level motor and perceptual processes.

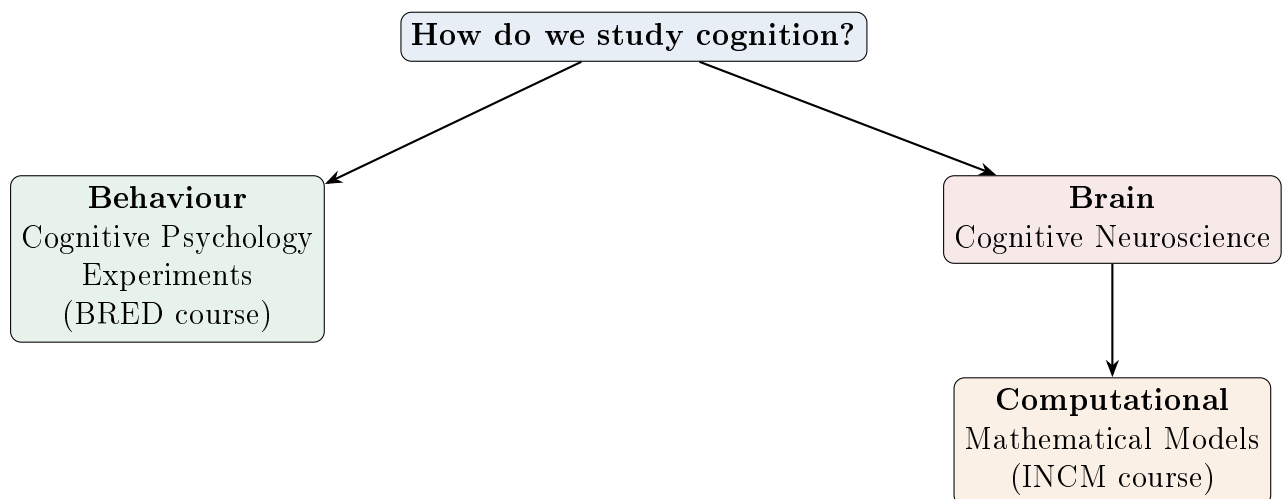
2.4 Brain: The Material Basis for Behaviour

Reductionist View

The **reductionist** position holds that everything we experience mentally has a **material basis in the brain**. Mind = brain activity. There is ongoing debate about whether consciousness can be fully reduced to neural activity.

Alternative view: Some traditions propose a non-material basis (“spirit” or “universal consciousness”) that animates the body. This is difficult to test empirically. As scientists, we proceed with the assumption of material causation while acknowledging the open question.

2.5 How Do We Study Cognition?



Three pillars of investigation:

- I. **Behaviour** — Cognitive psychology experiments (reaction time, accuracy)
- II. **Brain** — Cognitive neuroscience (neuroimaging, patient studies)
- III. **Computational Models** — Mathematical/computational models that bridge theory and experiment

2.5.1 Behavioural Experiments

- Conducted in controlled lab settings
- **Reaction time (RT)** tasks: present a stimulus, measure time to button press
- Average human reaction time $\approx$ **100 ms** (visual stimulus to motor response)
- Cricket batsmen need $\sim$ 15–50 ms response time—far below baseline RT; achieved through extensive training and anticipation
- Physical limitation: neural signal must travel from retina $\rightarrow$ visual cortex $\rightarrow$ motor cortex $\rightarrow$ spinal cord $\rightarrow$ muscles (electrochemical propagation is slow)

Additional tools: eye trackers, VR-based environments, physiological measurements (heart rate, skin conductance).

2.5.2 Studying the Brain: Healthy Participants

Non-invasive techniques:

- **EEG** (Electroencephalography) — electrodes on scalp measure electrical activity
- **MEG** (Magnetoencephalography) — measures magnetic fields from neural activity
- **fMRI** (functional MRI) — measures blood oxygenation changes (BOLD signal)

Invasive techniques:

- **PET** (Positron Emission Tomography) — inject radioactive tracer; measure glucose metabolism
- **ECoG** (Electrocorticography) — electrodes placed directly on the cortex (during surgery)

2.5.3 Studying the Brain: Patients and Neurological Disorders

Learning from Brain Damage

Much of what we know about brain–behaviour relationships comes from studying patients with brain injuries or neurodegenerative diseases. If area X is damaged and function Y is lost, we infer that area X participates in function Y .

Phineas Gage (1848)

A railroad worker had an iron rod shot through his **frontal lobe** (orbitofrontal cortex). He survived and was conscious, but his personality changed dramatically: he became impulsive, angry, socially inappropriate, and unable to plan. He lost his job and could not hold another.

Inference: The **frontal lobe** is critical for planning, social behaviour, personality, and emotional regulation—not vital for basic survival functions (breathing, heart rate) but essential for what makes us “human.”

Broca’s Aphasia

Paul Broca’s patient could understand speech (follow commands like “pour water into the glass”) but could *not produce* speech.

Affected area: **Broca’s area** in the **frontal lobe**.

Type: Expressive (non-fluent) aphasia.

Key inference: Speech production and speech comprehension are *independent modules* served by different brain regions.

Wernicke’s Aphasia

Carl Wernicke’s patient could speak fluently but produced nonsensical speech and could *not comprehend* language.

Affected area: **Wernicke’s area** in the **temporal lobe**.

Type: Receptive (fluent) aphasia.

Key inference: Confirms the **dissociation** between speech production (frontal) and comprehension (temporal).

Capgras Syndrome

Patients *recognise* their loved ones visually (“he looks like my husband, sounds like my husband”) but *believe* they are impostors. They claim the real person has been abducted and replaced.

Analysis — What is intact:

- Visual recognition (pattern matching) — intact
- Memory retrieval and comparison — intact

What is lost:

- **Emotional evaluation** — the “warm feeling” or emotional connection to the recognised face

Inference — Two systems:

1. A system for pattern processing, memory retrieval, and comparison
2. A separate system for **emotional valuation** (involving the **amygdala**)

These two systems are integrated in the **frontal cortex**. Damage to the amygdala, the emotional pathway, or the integrating region causes Capgras syndrome.

Other conditions studied:

- **Alzheimer’s dementia** — starts in temporal lobe; manifests as memory loss, disorientation
- **Frontotemporal dementia** — personality changes, impulsive behaviour, speech difficulties
- **Lobotomy** — historical (discredited) practice of removing frontal lobe tissue to treat personality disorders

2.5.4 The Role of Computational Modelling

Why Models Are Essential

1. **Manipulation** — in a model, you can switch components on/off and observe effects (impossible in vivo without waiting for specific diseases)
2. **Integration** — models combine observations from behaviour, neuroimaging, and patient studies into a unified framework
3. **Theory formation** — a complex system *cannot* be understood without a formal theory. No matter how sophisticated your measurements, without a model predicting how measurements relate, data remains uninterpretable.
4. **Theory–Experiment cycle** — model → prediction → experiment → refine model → new prediction → ...

This is exactly how physics works: cosmology does not work by observation alone; it requires theory.

Practice & Review

Q1. [Multiple Choice] The ability to focus on one conversation in a noisy party while ignoring all others is called:

- (A) The cocktail party effect (B) Divided attention (C) Qualia (D) Dual-task processing

Answer: A. The cocktail party effect is a classic demonstration of *selective (focused) attention*—the ability to prioritise one stimulus stream and suppress others.

Q2. [Short Answer] Define qualia. Why does their existence make consciousness scientifically challenging to study?

Answer: Qualia are the subjective, personal qualities of conscious experiences—e.g., the precise “crunchiness” and “sweetness” *you* experience biting an apple. Your qualia are not identical to anyone else’s and cannot be directly observed or verified by a third party. Because consciousness is inherently first-person and qualia cannot be measured objectively, standard scientific methods (which require repeatability and third-party verification) cannot easily access them.

Q3. [Essay] Compare EEG, fMRI, and PET as tools for studying the brain. For each, state what it measures, whether it is invasive, and its temporal/spatial resolution trade-off.

Answer: **EEG**—measures aggregate electrical activity of neurons via scalp electrodes (non-invasive); excellent *temporal* resolution ($\sim$ ms) but poor spatial resolution. **fMRI**—measures the BOLD signal (blood oxygenation level-dependent, an indirect proxy for neural activity, non-invasive); excellent *spatial* resolution (mm) but poor temporal resolution ($\sim$ seconds). **PET**—measures glucose metabolism via injected radioactive tracer (invasive); poor temporal resolution (minutes); largely

replaced by fMRI for most research since ~2000. Trade-off summary: EEG = when + what happened; fMRI/PET = where it happened.

- Q4. [Short Answer]** Describe the Capgras syndrome case. What two dissociated brain systems does it reveal?

Answer: Capgras patients visually recognise their loved ones (“He looks and sounds exactly like my husband”) yet believe them to be imposters—they insist the real person has been replaced. What is *intact*: (1) visual pattern recognition and memory comparison. What is *lost*: (2) emotional valuation—the “warm feeling” of familiarity (mediated by the amygdala and its connection to the frontal cortex). This reveals two independent systems: a cognitive/recognition system and a separate affective/emotional evaluation system, normally integrated in the frontal cortex.

- Q5. [Multiple Choice]** Broca’s area damage causes a patient to:

(A) Speak fluently but produce nonsensical language (B) Understand speech but be unable to produce speech
(C) Be unable to understand speech but produce fluent output (D) Lose all language ability entirely

Answer: **B**. Broca’s aphasia (expressive/non-fluent aphasia) results from damage to Broca’s area in the frontal lobe. Speech comprehension (Wernicke’s area, temporal lobe) remains intact.

- Q6. [Short Answer]** What happened to Phineas Gage and what did it demonstrate about the frontal lobe?

Answer: In 1848, a railroad worker survived an iron rod blasting through his orbitofrontal cortex. Although physically functional, his personality changed dramatically: he became impulsive, irritable, unreliable, socially inappropriate, and unable to plan or hold a job. His survival functions (breathing, heart rate) were unaffected. The case demonstrated that the frontal lobe is critical for **executive function** (planning, inhibitory control), **social behaviour**, and **personality**—the very functions that define us as human.

- Q7. [Short Answer]** Name the three pillars of cognitive science investigation introduced in this lecture. Which uses the BOLD signal?

Answer: (I) **Behaviour**—cognitive psychology experiments (reaction time, accuracy); (II) **Brain**—cognitive neuroscience (neuroimaging, patient studies); (III) **Computational models**—mathematical models linking theory to experiment. The BOLD (Blood Oxygenation Level-Dependent) signal is used by **fMRI** in pillar II.

- Q8. [Essay]** Why is computational modelling considered essential in cognitive neuroscience, not merely supplementary?

Answer: A complex system cannot be understood from measurement alone—a formal theory is required to *predict* how measurements should relate to each other. Modelling provides: (1) **Manipulation**—in a model you can selectively disable

components (impossible in a living brain without waiting for specific disease); (2) **Integration**—models synthesise findings from behaviour, imaging, and patient studies into a unified framework; (3) **Theory formation**—generates testable predictions; (4) **Theory–experiment cycle**—model $\rightarrow$ prediction $\rightarrow$ experiment $\rightarrow$ model refinement. Analogous to how physics could not advance without mathematical theory (e.g., cosmology without General Relativity).

Lecture 3: Neuroanatomy — Brain Structure (March 10)

3.1 Overview of the Brain

The brain is:

- Divided into **two hemispheres** (left and right), connected by dense fibres (e.g., **corpus callosum**)
- Composed of the **cerebrum**, **cerebellum**, and **brainstem/spinal cord**
- The cerebrum is also called the **neocortex** (“new cortex”)
- Protected by the **skull**, surrounded by **meninges** (including pia mater)

3.2 Grey Matter and White Matter

Grey Matter

Grey matter consists of neuronal **cell bodies** (soma). It appears darker. Found on the *outer surface* of the cerebral cortex and in the *interior* of the spinal cord.

White Matter

White matter consists of **myelinated axonal fibres** that connect neurons across regions. The myelin sheath (a fatty coating) gives it its white appearance. Found *inside* the cerebrum and on the *outside* of the spinal cord.

Key Contrast: Cortex vs. Spinal Cord

- **Cortex:** grey matter (outside), white matter (inside)
- **Spinal cord:** grey matter (inside, butterfly-shaped), white matter (outside)

This is an *opposite* arrangement!

An MRI scan reveals four tissue types: grey matter, white matter, cerebrospinal fluid (CSF), and bone.

3.3 Cerebrospinal Fluid (CSF)

Cerebrospinal Fluid

CSF is a clear, colourless fluid that surrounds the brain and spinal cord. It:

- Provides cushioning and mechanical protection
- Supplies nutrients
- Removes waste products
- Circulates through ventricles and channels within the brain

3.4 Central and Peripheral Nervous Systems

Central Nervous System (CNS)

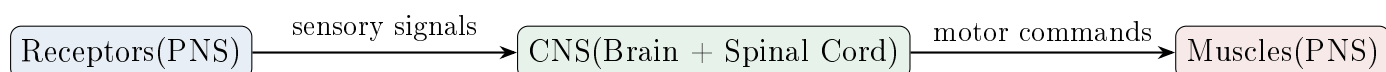
The **CNS** comprises the **brain** (cerebrum + cerebellum) and the **spinal cord**. It receives, processes, and integrates information, and generates output commands.

Peripheral Nervous System (PNS)

The **PNS** consists of all the nerves outside the CNS. It includes:

- **Sensory neurons** — collect information from receptors (touch, vision, hearing, smell, taste) and send it to the CNS
- **Motor neurons** — transmit commands from the CNS to muscles and glands

Sensory–Motor loop:



Example: Touch receptors on the skin sense mechanical distortion → electrical signals travel via axons to spinal cord → processed in brain → motor commands sent down spinal cord → muscles contract/relax (e.g., biceps contracts + triceps relaxes → arm bends at elbow).

3.5 Brain Sections (Planes of Cut)

Section	Plane	Shows
Coronal	Front-to-back cut	Both hemispheres (L & R)
Horizontal (Transverse/Axial)	Top-to-bottom cut	Both hemispheres (L & R)
Mid-Sagittal	Left-right midline cut	One hemisphere only

Directional terminology:

- **Anterior** (front) / **Posterior** (back)
- **Superior/Rostral** (top) / **Inferior/Caudal** (bottom)
- **Ventral** (front side in humans) / **Dorsal** (back side in humans)
- (Note: these terms originate from quadrupedal animals, causing some confusing mapping to bipedal humans)

3.6 Cortical Surface Features

Gyrus, Sulcus, Fissure

- **Gyrus** (pl. gyri) — a “bump” or ridge on the cortical surface
- **Sulcus** (pl. sulci) — a “valley” or groove between gyri
- **Fissure** — a deep, prominent groove (larger than a sulcus)

The cortex is folded (convoluted) to pack more surface area into the skull.

Major landmarks:

Landmark	Separates
Central Sulcus	Frontal lobe (anterior) from Parietal lobe (posterior)
Lateral (Sylvian) Fissure	Temporal lobe (inferior) from Frontal/Parietal (superior)
Longitudinal Fissure	Left hemisphere from Right hemisphere
Transverse Fissure	Cerebrum from Cerebellum

3.7 The Four Lobes of the Cerebrum

3.7.1 (1) Frontal Lobe

Located **anterior** to the central sulcus. The **newest** part of the neocortex; most developed in humans.

Frontal Lobe Functions

- **Executive function:** working memory, planning, decision-making, inhibitory control
- **Motor cortex:** strip just anterior to central sulcus; controls all voluntary motor actions
- **Speech production:** Broca’s area (in frontal lobe)
- **Emotional evaluation,** social behaviour, personality
- Continues developing until age ~**21 years**

Executive function includes:

- **Working memory** — temporary scratch-pad for ongoing tasks
- **Inhibitory control** — “go/no-go”; knowing when *not* to act (develops in children; why infants put fingers in electrical sockets)
- **Planning** — sequencing actions (e.g., making a shopping list → take money → take bag → go to market → buy items); long-horizon planning (career, retirement)

“Out of Sight, Out of Mind” — Working Memory Development

Object permanence experiment: A toy is shown to an infant and moved behind a screen.

- A ~5-month infant: loses interest once the toy goes behind the screen (no working memory persistence)
- A ~6-month infant: tracks and expects the toy to emerge on the other side

Working memory capacity increases with age, reaching adult levels around 8–9 years.

Phineas Gage revisited: Damage to frontal lobe → loss of planning, social behaviour, personality change—but basic survival functions remained intact. Frontal lobe is “not vital for breathing but essential for being human.”

3.7.2 (2) Parietal Lobe

Located **posterior** to the central sulcus.

Parietal Lobe Functions

- **Somatosensory cortex:** strip just posterior to central sulcus; receives sensation from all over the body
- **Proprioception:** sense of where body parts are in space without looking (close eyes → know your hand is raised)
- **Spatial orientation:** ego-centric (relative to self) and allocentric/world-centred (relative to other objects)
- **Goal-directed action:** guides motor actions to spatial targets (e.g., swatting a mosquito on your arm)

Vestibular system: Sensors in the inner ear detect orientation/balance. Ear infections can cause vertigo (“world is moving”) because the sensor provides incorrect orientation data.

3.7.3 (3) Occipital Lobe

Located at the **back** of the brain, above the transverse fissure.

Occipital Lobe Functions

- **Visual cortex:** primary centre for all visual processing
- Object recognition and identification
- Colour, shape, and size perception
- Object motion detection (direction, velocity)
- Two visual streams: one for **object recognition** (“what”), one for **motion/spatial** (“where/how”)

3.7.4 (4) Temporal Lobe

Located beneath the lateral (Sylvian) fissure, under the temporal bone.

Temporal Lobe Functions

- **Auditory cortex:** processing all sounds
- **Speech comprehension:** Wernicke’s area (in temporal lobe)
- **Language processing**
- **Object naming:** associating labels (“chair”, “table”) with recognised objects
- Input from ears → hair cells convert pressure waves to electrical signals → auditory cortex

3.7.5 (5) Insula (“Hidden” Fifth Lobe)

Tucked inside the temporal lobe, not visible from the outside.

- **Sense of self**
- **Saliency processing** — which stimuli are more/less salient

3.8 Subcortical Structures

Clusters of neurons located *beneath* the cortex. These are phylogenetically **older** structures, present even in simpler animals.

Structure	Function
Amygdala	Fear processing; emotional response to threatening stimuli
Hippocampus	Long-term memory formation (horseshoe-shaped)
Basal Ganglia	Motor control; gateway for motor output to cortex
Thalamus	Sensory relay station; gathers incoming sensory info & distributes to cortex
Hypothalamus	Hunger, thirst, temperature regulation, breathing, rage
Cerebellum	Motor coordination; balance; motor learning

Thalamus and Basal Ganglia as Gateways

Thalamus = gateway for *sensory* information **into** the cortex.

Basal Ganglia = gateway for *motor* commands **out of** the cortex.

Cerebellum and Alcohol

Long-term alcohol abuse **compromises the cerebellum**, leading to loss of motor coordination (swaying, inability to walk straight). This is also why police use the “walk in a straight line” test for intoxication—alcohol temporarily shuts down cerebellar signals. The condition is called **cerebellar ataxia**.

3.9 The Midbrain and Brainstem (Old Brain)

The **midbrain/brainstem** controls the most vital functions:

- Breathing, respiration
- Heart rate regulation
- Sleep–wake cycle
- Temperature control

If You Had to Lose Part of Your Brain...

Given a choice, one would rather lose the **cerebrum** (neocortex) than the **midbrain**. The midbrain controls survival-critical functions. Many motorcycle accident victims enter **coma** because the brainstem (where breathing circuits reside) is damaged by mechanical shock.

Wear a helmet—it lessens shock to the brain, especially the brainstem–spinal cord junction.

3.10 Brain Development

- The CNS is **not** fully developed at birth
- Full maturation continues until approximately **age 21**
- Developmental milestones: visual system set by ~1 year, auditory/speech by ~1.5–2 years, frontal lobe by ~18–21 years
- Development involves *growing connections* (not growing new brain tissue)—inactive connections become active through experience
- Humans have the **longest post-birth gestation period** of any animal: rolling → raising head → crawling → standing → walking → speech → sense of self

Mirror Test for Self-Awareness

A chimpanzee with a dot on its forehead was placed in front of a mirror. It tried to wipe the dot *off the mirror* rather than off its own forehead—indicating a lack of self-recognition. In human infants, the sense of self emerges gradually during development.

Practice & Review

Q1. [Multiple Choice] Grey matter in the brain primarily consists of:

- (A) Myelinated axonal fibres (B) Cerebrospinal fluid (C) Neuron cell bodies (soma) (D) Synaptic vesicles

Answer: C. Grey matter is composed of neuronal cell bodies and their supporting glial cells. It appears darker in colour. White matter consists of myelinated axons (myelin gives the white appearance).

Q2. [Short Answer] What is the key difference in the arrangement of grey and white matter between the cerebral cortex and the spinal cord?

Answer: In the **cerebral cortex**: grey matter is on the *outside* (cortical surface), white matter is on the *inside*. In the **spinal cord**: the arrangement is *reversed*—grey matter forms a butterfly-shaped core on the *inside*; white matter surrounds it on the *outside*. This opposite arrangement is a frequently examined fact.

Q3. [Short Answer] Name the four lobes of the cerebrum and give one primary function of each.

Answer: (1) **Frontal lobe**—executive function, motor control (motor cortex), speech production (Broca’s area), planning, personality; (2) **Parietal lobe**—somatosensory processing, proprioception, spatial orientation; (3) **Occipital lobe**—visual processing (primary visual cortex); (4) **Temporal lobe**—auditory processing, speech comprehension (Wernicke’s area), object naming.

Q4. [Multiple Choice] The groove (valley) between two gyri on the cortical surface is called a:

- (A) Sulcus (B) Gyrus (C) Fissure (D) Ventricle

Answer: A. A *sulcus* is a groove; a *gyrus* is a ridge/bump. A *fissure* is a deep, major groove (e.g., the lateral/Sylvian fissure separating the temporal lobe from the frontal/parietal lobes).

Q5. [Essay] Describe the frontal lobe’s role in executive function. What are its three components? Use the Phineas Gage case and the “object permanence” experiment to illustrate its development.

Answer: Executive function has three components: (1) **Working memory**—temporary scratch-pad for ongoing tasks; (2) **Inhibitory control**—“go/no-go” responses; knowing when *not* to act (reason infants put fingers in sockets—frontal lobe not yet mature); (3) **Planning**—sequencing actions over time (shopping list, career planning). **Phineas Gage**: iron rod through orbitofrontal cortex →

lost planning, social behaviour, and personality while survival functions remained intact—frontal lobe is “not vital for breathing but essential for being human.”

Object permanence: a ~5-month infant loses interest when a toy disappears behind a screen (no working memory persistence); a ~6-month infant tracks and waits for it (working memory emerging). Frontal lobe matures until ~21 years.

- Q6. [Short Answer]** Contrast the roles of the thalamus and the basal ganglia. Use the “gateway” metaphor.

Answer: **Thalamus** = *sensory gateway into the cortex*—collects all incoming sensory information and distributes it to the appropriate cortical areas. **Basal ganglia** = *motor gateway out of the cortex*—a relay for motor output commands. They operate in opposite directions: thalamus (in), basal ganglia (out).

- Q7. [Multiple Choice]** Long-term alcohol abuse damages the cerebellum, leading to:
(A) Memory loss (Korsakoff syndrome) (B) Cerebellar ataxia (loss of motor coordination)
(C) Inability to speak (D) Complete personality change

Answer: **B.** Chronic alcohol compromises the cerebellum, causing *cerebellar ataxia*—swaying gait, inability to walk in a straight line, loss of balance and coordination. This is the basis of the police “walk the line” sobriety test.

- Q8. [Short Answer]** List three functions of cerebrospinal fluid (CSF) and describe where it is found.

Answer: CSF is a clear, colourless fluid that surrounds the brain and spinal cord (circulating through ventricles and channels). Functions: (1) **Mechanical cushioning**—protects the brain from physical impact; (2) **Nutrient delivery**—supplies oxygen and nutrients to CNS tissue; (3) **Waste removal**—clears metabolic by-products; (4) **Pressure regulation**—maintains stable intracranial pressure.

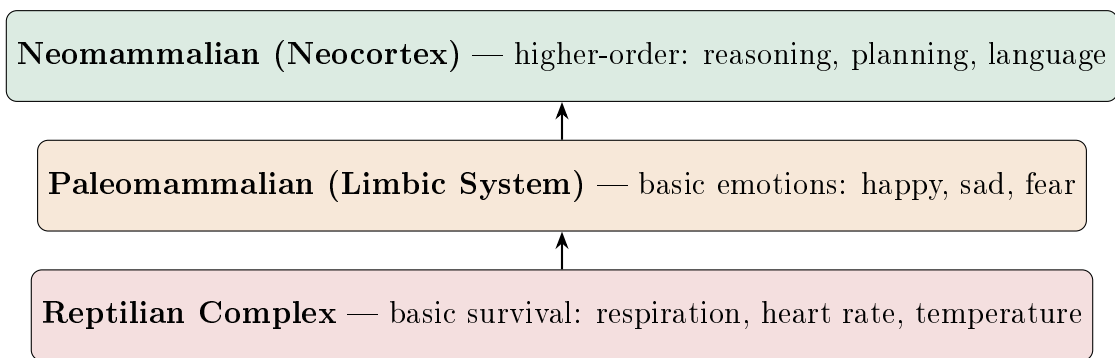
- Q9. [Short Answer]** What is proprioception and which lobe handles it? Give an example of proprioception in action.

Answer: Proprioception is the sense of the position and orientation of one’s own body parts in space *without looking*. It is processed in the **parietal lobe** (somatosensory cortex). Example: close your eyes and raise your right hand—you know where your hand is without visual feedback. Proprioceptive signals come from receptors in muscles, tendons, and joints.

Lecture 4: Brain Imaging, Neurons, and Scales (March 13)

4.1 Brain Evolution: The Triune Brain

The **triune brain theory** (Paul MacLean) proposes three evolutionary layers:



- The older layers are **co-opted** from evolutionary ancestors (shared with reptiles, lower mammals)
- The **prefrontal cortex** is the most developed in humans—supports planning, decision-making, emotional regulation, higher-order thinking
- Basic functions (respiration, hunger, temperature) are common across species
- Studying reptiles and mammals helps us understand our own brain
- Evolutionary trade-offs: e.g., dogs have superior smell but inferior visual processing compared to humans

4.2 Resting State Brain Function

Resting State

Resting state refers to brain activity when a person is *not* performing any specific task (e.g., lying in an MRI scanner with eyes closed). The brain remains highly active even at “rest.”

- Resting state brain connectivity differs between healthy individuals and those with:
 - Alzheimer’s disease
 - Dementia
 - Coma / vegetative state
- This has become a powerful diagnostic tool for neurological and psychiatric conditions

- **Application:** predicting recovery in coma patients from resting-state fMRI patterns (still an open research problem)
- The **brain operates 24/7/365**—if it stops, you are dead

4.3 Brain Imaging Techniques (Detailed)

4.3.1 Positron Emission Tomography (PET)

PET Scan

PET measures **glucose metabolism** in the brain. A radioactive tracer (radioisotope) is injected into the bloodstream, travels to the brain, and as it decays, emits positrons that are detected by the scanner.

- Active brain regions consume more glucose → higher metabolism → “hot” colours (red, yellow)
- **Example:** Eyes closed (low occipital activity) → Eyes open (occipital cortex activates) → Complex visual scene (further increased visual cortex activity)
- **Invasive:** requires injection of radioactive substance
- Older technique (1980s–90s); less commonly used today

4.3.2 Structural MRI

- Uses magnetic resonance to image brain *structure*
- Shows grey matter, white matter, CSF, and bone with different contrasts
- Used to identify structural damage (tumours, lesions, atrophy)
- Three views: coronal, horizontal/axial, sagittal

4.3.3 Electroencephalography (EEG)

EEG

EEG measures the aggregate **electrical activity** of neurons from electrodes placed on the scalp. It captures the synchronised firing of ~10,000+ neurons in a region.

EEG frequency bands and brain states:

State	Frequency	Amplitude
Deep sleep	Low (delta band)	High amplitude
Drowsy (pre-sleep)	Medium (alpha band)	Medium
Awake & alert	High (beta/gamma band)	Low amplitude

EEG Signatures

- **Awake:** high frequency, low amplitude — neurons firing independently and rapidly, exchanging information
- **Deep sleep:** low frequency, high amplitude — neurons firing synchronously in slow waves
- Frequency bands: **delta** (lowest) < **theta** < **alpha** < **beta** < **gamma** (highest)

Event-Related Potentials (ERPs):

- Present a stimulus (e.g., auditory tone) and measure the brain's electrical response
- Time-locked to stimulus onset; averaged over many trials
- Signal amplitude: $\sim 2 \mu\text{V}$; duration: ~ 700 ms post-stimulus
- ERP components reveal stages of information processing

EEG systems: 8, 16, 32, 64, 128, or 256 electrodes. More electrodes = denser spatial sampling. Electrodes arranged using the **10–20 system** (every 20% of head distance).

MEG (Magnetoencephalography): Measures the *magnetic field* generated by the same electrical activity that EEG detects. Requires very sensitive sensors (SQUIDS).

4.3.4 Functional MRI (fMRI)

fMRI

fMRI measures the **Blood Oxygenation Level-Dependent (BOLD)** signal. When a brain region is active, it demands more oxygenated blood. The ratio of oxygenated to deoxygenated haemoglobin changes the local magnetic signal, which the MRI detects.

PET vs. fMRI

- **PET:** measures glucose *metabolism*; invasive (radioactive tracer)
- **fMRI:** measures blood *oxygenation* changes; non-invasive
- Both are *indirect* measures of neural activity
- fMRI breakthrough: ~ 1999 – 2000 ; transformed cognitive neuroscience by allowing functional studies without invasive procedures

Key fMRI finding — Mental imagery:

- When participants *view* pictures: strong activity in visual cortex (occipital lobe)
- When participants *imagine* objects (no physical stimulus): visual cortex still activates, but with *reduced* intensity

- Higher-order visual areas show similar activity for real and imagined stimuli
- Same result for motor imagery: imagining a hand movement activates motor cortex (less intensely than actual movement)

4.4 Brain–Computer Interfaces (BCIs)

Brain–Computer Interface

A **BCI** decodes brain activity (e.g., motor intention from EEG/ECoG) and interfaces it with a computer or robotic device, enabling control through thought alone.

Application: Patients with paralysis (e.g., stroke, ALS/motor neuron disease like Stephen Hawking) who can still *think* about movement can have their motor intentions decoded and used to control a robotic arm, generate speech, or type.

4.5 Deep Brain Stimulation (DBS)

Deep Brain Stimulation

DBS involves surgically implanting electrodes into specific subcortical structures (e.g., basal ganglia) and delivering a mild electrical current to modulate neural activity.

- Primary application: **Parkinson’s disease** (a motor disorder with tremors and rigidity)
- When DBS is switched **on**: symptoms dramatically reduce; patient can move normally
- When DBS is switched **off**: symptoms gradually return
- Used when medication is insufficient in severe cases

4.6 The Neuron

Neuron

A **neuron** is the fundamental computational unit of the nervous system. It is an **electrically excitable** cell with three main parts:

1. **Dendrites** — receive input from other neurons
2. **Cell body (soma)** — aggregates inputs spatially and temporally
3. **Axon** — generates and propagates the output (**action potential**) to other neurons

Key properties:

- Action potential amplitude: ~ 30 mV (from resting ~ -70 mV)
- Average connectivity: each neuron connects to $\sim 10^4$ (10,000) other neurons
- Signal propagation is **regenerative**: the action potential is regenerated at full amplitude at each node, not split/attenuated (unlike an electrical wire splitting current among branches)
- The **myelin sheath** (fatty coating on axons) gives rise to white matter

4.7 Synapses

Synapse

A **synapse** is the junction between an axon terminal and the dendrite of the next neuron. The gap ($\sim 1 \mu\text{m}$) is called the **synaptic cleft**. Communication is **electro-chemical**:

1. Electrical signal (action potential) arrives at axon terminal
2. Chemical **neurotransmitters** are released into the synaptic cleft
3. Neurotransmitters bind to receptors on the post-synaptic dendrite
4. A new electrical signal is generated in the receiving neuron

This electrochemical process makes neural communication **slow** compared to electronic circuits.

4.8 Spatial Scales of the Nervous System

Scale	Size	Structure
Molecular	Ångströms (10^{-10} m)	Ion channels, neurotransmitters
Synaptic	Micrometres (μm)	Synaptic cleft, vesicles
Neuronal	$\sim 100 \mu\text{m}$	Single neuron cell body
Networks	Millimetres	Local circuits of connected neurons
Maps	Centimetres	Motor map, somatosensory map
Systems	~ 10 cm	Visual system, auditory system
CNS	~ 1 m	Entire brain + spinal cord

4.9 Temporal Scales of Cognitive Phenomena

Time Scale	Order	Phenomenon
10^{-3} s (1 ms)	Milliseconds	Action potential firing
10^{-1} s (100 ms)	Hundred ms	Object recognition; simple reaction time
10^0 s (1 s)	Seconds	Simple motor action
10^1 s (10 s)	Tens of seconds	Habituation
10^2 s (100 s)	Minutes	Planning a chess move
10^5 – 10^6 s	Days–Months	Perceptual learning; skill acquisition (e.g., cycling)
10^7 s	Months–Years	Learning to walk (infant)

4.10 Memory and Learning at the Synapse

Synaptic Plasticity

Learning occurs by **increasing the strength of synaptic connections**. All memory is encoded in connection strengths. In some cases, previously inactive connections become active through repeated experience.

This is the neural basis of the **nature vs. nurture** debate: the “wiring” is in place (nature), but connections are refined through experience (nurture). Both contribute to cognitive development.

Nobel Prize (2000): Awarded for discovery of synaptic changes due to learning.

Nobel Prize (1957): Awarded for discovery of action potential mechanisms.

4.11 Brain vs. Digital Computer

Feature	Digital Computer	Brain
Processing unit	Complex CPU (ALU); GHz speed	Simple neuron; ms speed
Number of processors	1–thousands	$\sim 10^{10}$ (billions)
Processing style	Sequential (mostly)	Massively parallel
Speed	$\sim 10^9$ operations/s	$\sim$ ms per operation (slow)
Communication	Electrical (fast)	Electrochemical (slow)
Memory architecture	Separate (RAM, HDD, cache); localised	Distributed ; memory at every synapse
Memory addressing	Address-based (location)	Content-addressable
Learning	Stored program	Self-learning
Robustness	Vulnerable (heat, damage)	Robust (works at 38°C+; tolerates injury)
Reliability	100% deterministic	Probabilistic; graceful degradation
Data type	Numerical & symbolic	Conceptual
Operating environment	Well-defined, constrained	Unconstrained, real-world

Content-Addressable Memory

In a computer, memory is accessed by **address** (location). In the brain, memory is **content-addressable**: you recall your friend's face by thinking about them, not by sending an address to a specific memory location. The memory is distributed across synapses throughout the brain, not stored in one place.

Practice & Review

Q1. [Multiple Choice] fMRI primarily measures which of the following to infer brain activity?

- (A) Electrical impulses on the scalp (B) Changes in blood oxygenation (BOLD signal)
 (C) Glucose metabolism via radiotracer (D) Magnetic fields from ion channels

Answer: B. fMRI detects the Blood Oxygenation Level-Dependent (BOLD) signal. Active brain regions demand more oxygenated blood; the ratio of oxy- to deoxy-haemoglobin changes the local MRI signal. It is non-invasive and replaced PET as the dominant functional imaging method after ~ 2000 .

Q2. [Short Answer] Describe the triune brain theory. Name the three layers from oldest to newest and identify which part is most developed in humans.

Answer: The triune brain (Paul MacLean) proposes three evolutionary layers: (1) **Reptilian complex** (brainstem)—basic survival: respiration, heart rate, temperature; (2) **Paleomammalian brain** (limbic system)—basic emotions: fear, happiness, anger; (3) **Neomammalian brain** (neocortex)—higher-order reasoning, planning, language. The **prefrontal cortex** (part of the neocortex) is most developed in humans and supports our most advanced cognitive capacities.

Q3. [Essay] Explain brain–computer interfaces (BCIs). Describe two types, their invasiveness, and a clinical application for each.

Answer: A BCI decodes brain activity (motor intention, thoughts) and interfaces it with a computer or robotic device, enabling control through thought alone. (1) **Surface BCI** (non-invasive): EEG electrodes on the scalp decode motor intent. Application: IIIT research decoding motor intention from EEG for stroke/hemiplegia patients to drive physiotherapy robots. Limitation: lower signal-to-noise. (2) **Implanted BCI** (invasive): electrodes implanted in brain tissue (e.g., Neuralink). Application: enabling paralysed patients (ALS, spinal cord injury) to control robotic arms, type, or communicate. Higher precision but requires surgery with associated infection risk. Both types demonstrate the power of understanding motor intention circuitry.

Q4. [Short Answer] What is synaptic plasticity? Connect it to memory and to the nature vs. nurture debate.

Answer: Synaptic plasticity is the ability of synaptic connections to strengthen or weaken based on activity patterns (e.g., Hebbian learning: “neurons that fire together, wire together”). All memory is encoded in synaptic connection strengths—learning = selectively strengthening relevant connections through repeated experience (Nobel Prize 2000). Nature–nurture connection: the neural “wiring” (number of neurons, basic connectivity) is genetically determined (nature), but which specific connections become strong or weak depends on experience (nurture). Both contribute.

Q5. [Multiple Choice] In EEG recordings, a state of deep sleep is characterised by:
 (A) High frequency, low amplitude activity (B) Low frequency, high amplitude activity
 (C) Complete electrical silence (D) High frequency, high amplitude activity

Answer: **B.** Deep sleep = delta band (lowest frequency, highest amplitude); neurons fire synchronously in slow waves. Being awake and alert = beta/gamma band (high frequency, low amplitude); neurons fire independently and rapidly, exchanging information. The inverse relationship of frequency and amplitude is a key EEG principle.

Q6. [Short Answer] Name the five EEG frequency bands from lowest to highest and give the associated brain state for each extreme.

Answer: **Delta** (0.5–4 Hz): deep sleep; **Theta** (4–8 Hz): drowsy, light sleep; **Alpha** (8–13 Hz): relaxed but awake, eyes closed; **Beta** (13–30 Hz): active, attentive; **Gamma** (>30 Hz): highly focused processing, problem-solving. The 10–20 electrode placement system spaces electrodes every 20% of head distance.

- Q7. [Short Answer]** Explain Deep Brain Stimulation (DBS) for Parkinson’s disease. What structure is targeted and what is the evidence for its effectiveness?

Answer: DBS involves surgically implanting an electrode into the **subthalamic nucleus** (part of the basal ganglia) and delivering a mild electrical current. Parkinson’s disease involves dysfunction of the basal ganglia leading to resting tremor and rigidity. Evidence: when DBS is switched **on**, the patient’s tremor dramatically reduces within seconds and they can move normally; when switched **off**, tremors gradually return over minutes. The patient carries a switch. Used for advanced Parkinson’s when medication is insufficient. DBS demonstrates the precise role of the subthalamic nucleus in motor control.

- Q8. [Essay]** Compare the brain with a digital computer across five dimensions. Which comparison is most surprising and why?

Answer: Key comparisons: (1) **Processing:** computer = sequential; brain = massively parallel ($\sim 10^{10}$ neurons). (2) **Speed:** computer = GHz (10^9 ops/s); brain = ms per operation (slow), but compensates with parallelism. (3) **Memory:** computer = address-based (location); brain = **content-addressable** (recalled by content, distributed across synapses). (4) **Learning:** computer = stored program; brain = self-learning (synaptic plasticity). (5) **Robustness:** computer = brittle (single bit flip can crash); brain = robust, works at 38 °C+, tolerates neuron death with graceful degradation. Most surprising: content-addressable memory—recalling your friend by thinking about them, without knowing a “memory address.”

Lecture 5: Perception, Neural Coding, and Organisation Principles (March 17)

5.1 Perception

Perception

Perception is the interpretation of raw sensory input data. It is not passive recording but an active, constructive process by which the brain makes sense of incoming signals.

Visual illusions demonstrate that perception is not veridical (faithful to physical reality)—the brain constructs interpretations that can be “wrong” relative to the physical stimulus.

5.2 Neurons: Electrically Excitable Cells

Recap and extension:

- Neurons are **electrically excitable**—they respond to stimulation by generating electrical discharges (action potentials)
- **Axons carry information** from the cell body to distant targets
- Unlike other body cells, neurons have the unique property of electrical excitability

5.3 Synapses: Electrochemical Transmission

- The *majority* of synapses in the brain are **chemical synapses** (electrochemical transmission via neurotransmitters)
- A minority are **gap junctions** (direct electrical coupling), which have been **reduced evolutionarily**
- Chemical synapses are slower but allow for modulation, learning, and plasticity

5.4 Hodgkin–Huxley Model

Hodgkin–Huxley Model

The **Hodgkin–Huxley model** (Nobel Prize, 1963) is a mathematical model describing how action potentials are initiated and propagated in neurons. Developed using the **giant squid axon** as a model system.

Key voltage values:

- **Resting potential:** ≈ -70 mV
- **Threshold potential:** ≈ -55 mV (when exceeded, an action potential fires)

When the membrane potential depolarises from -70 mV past -55 mV, an **all-or-nothing** action potential is triggered ($\sim +30$ mV peak), propagates down the axon, and is regenerated at each node.

5.5 Inter-Pulse Intervals and Neural Coding

- Neurons encode information in the **timing** and **rate** of action potentials
- **Inter-pulse intervals** (time between successive spikes) carry information about stimulus intensity and other features
- Higher firing rates generally indicate stronger stimulation

5.6 From Action Potentials to Perception: Modularity

Modularity

Modularity is the principle that different cognitive functions are served by distinct, specialised brain modules (regions). The path from raw neural activity (action potentials) to rich perceptual experience involves hierarchical and modular processing.

5.7 Grandmother Coding vs. Distributed Processing

Grandmother Cell (Localist Theory)

The **grandmother cell hypothesis** proposes that specific individual neurons represent specific complex concepts (e.g., one neuron for your grandmother's face). This is a **localist** theory of representation.

Distributed Processing

Distributed (population) coding proposes that representations are spread across *many* neurons. No single neuron encodes a complete concept; instead, the pattern of activity across a population encodes the representation. This is also called **multi-modal distribution**.

Localist vs. Distributed

- **Grandmother cell:** one neuron = one concept. Simple but fragile (if that neuron dies, the concept is lost).
- **Distributed processing:** a pattern across many neurons = one concept. Robust (damage to some neurons causes graceful degradation, not total loss).
- Modern evidence supports **distributed processing** as the dominant coding scheme, though some neurons do show remarkable selectivity (e.g., “Jennifer Aniston neurons” in the medial temporal lobe).

Practice & Review

- Q1. [Multiple Choice]** The primary distinction between sensation and perception is:
- (A) Sensation is the processing and transmission of sensory input; perception is the brain's *interpretation* of that input
 - (B) Sensation is conscious; perception is unconscious
 - (C) Sensation occurs in the brain; perception in the peripheral nervous system
 - (D) They are synonymous terms for the same process

Answer: A. Sensation happens at the sensory organs and is conveyed to the CNS; perception is the constructive, interpretive process in the brain that creates our internal model of reality. Perception is not a faithful copy of the world—it is an active reconstruction (demonstrated by visual illusions).

- Q2. [Short Answer]** What is the Hodgkin–Huxley model? State the two critical voltage values and what happens at each.

Answer: The Hodgkin–Huxley model (Nobel Prize 1963, developed using the giant squid axon) mathematically describes how action potentials are initiated and propagated. (1) **Resting potential** ≈ -70 mV: the stable negative charge across the neuron membrane when at rest. (2) **Threshold potential** ≈ -55 mV: when depolarisation exceeds this, an *all-or-nothing* action potential fires, reaching $\approx +30$ mV peak, and propagates regeneratively down the axon.

- Q3. [Essay]** Compare the grandmother cell (localist) theory with distributed (population) coding. What does the evidence support? What are the implications for brain robustness?

Answer: **Grandmother cell hypothesis:** one specific neuron encodes one specific complex concept (e.g., your grandmother's face). Simple but fragile: if that neuron dies, the concept is irretrievably lost. **Distributed (population) coding:** a concept is encoded as a *pattern of activity* across many neurons; no single neuron holds the complete representation. Robust: losing a few neurons causes *graceful degradation* (partial loss of fidelity, not total loss). Modern evidence strongly supports distributed coding as the dominant scheme. However, highly selective neurons exist—"Jennifer Aniston neurons" in the medial temporal lobe respond almost exclusively to a specific person's face/name—suggesting a continuum rather than a binary choice. Robustness implication: the brain can tolerate neuronal death (e.g., stroke, Alzheimer's progression) with gradual decline rather than catastrophic failure.

- Q4. [Multiple Choice]** The Kanizsa triangle illusion (a white triangle perceived where none is drawn) demonstrates:
- (A) Top-down processing using prior knowledge
 - (B) Bottom-up processing driven by stimulus properties
 - (C) Working memory capacity limits
 - (D) Classical conditioning

Answer: B. Bottom-up processing: the visual system constructs contours from the arrangement of pac-man shapes, regardless of the viewer's language or cul-

tural knowledge. This is not driven by prior expectations.

- Q5. [Short Answer]** Explain the difference between top-down and bottom-up perceptual processing. Give one specific example of each from the lecture.

Answer: **Bottom-up:** stimulus properties directly drive perception without prior knowledge. Example: Kanizsa triangle (all viewers, regardless of language, perceive the triangle); café wall illusion (parallel lines appear sloped due to contrast). **Top-down:** stored knowledge, templates, and expectations guide interpretation of ambiguous stimuli. Example: reading a word printed with occluded letters (stripes over “IBC”)—your knowledge of English fills the gaps. A person who does not know English cannot read the occluded word.

- Q6. [Short Answer]** What is modularity in the context of brain organisation? Why is the modular arrangement efficient?

Answer: Modularity: different cognitive functions (vision, audition, motor control, language) are served by distinct, specialised brain regions (modules). Efficient because: (1) modules can process in *parallel*; (2) damage to one module selectively impairs that function without destroying others (selective deficits, e.g., Broca’s vs. Wernicke’s aphasia); (3) evolutionary refinement of one module does not require redesigning the whole system.

- Q7. [Multiple Choice]** The majority of synaptic connections in the brain use which type of transmission?

(A) Direct electrical (gap junctions) (B) Electrochemical (chemical synapses with neurotransmitters)
(C) Optical (photon-based) (D) Mechanical (pressure-based)

Answer: **B.** Chemical synapses are the dominant form; gap junctions have been *reduced* evolutionarily. Chemical synapses are slower but allow for modulation, learning (via neurotransmitter release modulation), and synaptic plasticity.

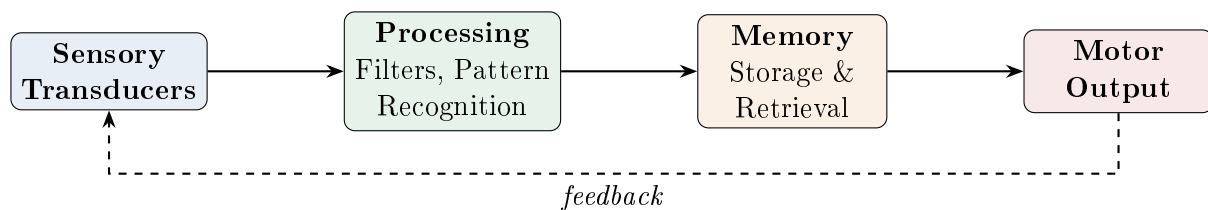
- Q8. [Essay]** What is the fusiform face area (FFA)? Connect it to distributed representation and to the Capgras syndrome case study.

Answer: The fusiform face area (FFA) is a region in the temporal lobe (fusiform gyrus) that is *specialised* for face processing—it responds strongly and selectively to faces, much more than to objects. Distributed representation link: although the FFA shows high selectivity for faces, full face recognition also engages the occipital lobe (early visual processing), parietal lobe (spatial features), and prefrontal cortex (decision about identity). Capgras link: Capgras patients have an intact FFA (they *recognise* the face) but the pathway to the amygdala (emotional evaluation) is damaged—so recognition is present but the emotional “familiarity” signal is absent. This confirms the two-route model: cognitive recognition vs. emotional evaluation.

Lecture 6: Cognitive Architecture, Memory Taxonomy, and Perception (March 24)

6.1 Architecture of Human Cognition

The brain can be understood as a **hierarchical input–output system**:



Sensory Transducers

Sensory transducers convert external physical signals (photons, pressure waves, chemical molecules, temperature) into the brain's internal electrical signals. Examples:

- **Retina** — converts photons → electrical signals
- **Cochlea (inner ear)** — converts pressure waves → electrical signals
- **Skin receptors** — convert mechanical/thermal stimuli → electrical signals

After transduction, **preliminary analysis** occurs: detecting brightness, edges, foreground vs. background, corners, depth—these are the **basic filters**. This information is then synthesised hierarchically into **patterns**.

The Brain Reconstructs Reality

The brain does not passively record the world. It **internally regenerates its own version of reality** from filtered, transduced signals. What you perceive is a *reconstruction*, not a direct copy of the physical world.

The cognitive architecture also includes:

- **Memory processes** — encoding, storage, and retrieval
- **Thought processes** — reasoning, inference, decision-making
- **Desires, motivations, intentions** — future goals, plans, sub-goal monitoring
- **Working memory** — temporary storage for ongoing tasks
- **Attentional mechanisms** — resource allocation in a parallel system
- **Emotions** — personalisation, sense of self, survival instincts
- **Consciousness/awareness** — self-awareness at the top of the hierarchy

6.2 Serial vs. Parallel Processing

Humans Are Parallel Processors

Unlike digital computers (primarily serial), humans process information **in parallel**. While speaking (serial output), we simultaneously:

- Make hand gestures (motor cortex)
- Monitor visual surroundings (visual cortex)
- Maintain balance (cerebellum)
- Breathe (brainstem)

If we were purely serial, we could not speak and breathe at the same time.

Attention is the mechanism that allocates resources in this parallel system—focusing on what is relevant while suppressing what is not.

6.3 Emotions in Cognition

Emotions

Emotions are complex psychological and physiological states that influence cognition, memory, and behaviour. Their role includes:

1. **Survival instinct** — primordial fear drives avoidance of danger (self-preservation across all biological systems)
2. **Emotionally-tagged memory** — emotionally significant events are stored more vividly and retrieved more easily
3. **Empathy** — the ability to simulate another person's emotional state ("I can put myself in your shoes")
4. **Sense of self** — emotions personalise experience; they define your likes, dislikes, values, and identity
5. **Approach/avoidance behaviour** — emotions convert neutral stimuli into personally significant ones ("I like this" vs. "I dislike this")

Open Question: Emotion–Cognition Link

The linkage between emotion and cognition is still not fully understood. Active research explores how emotions modulate cognitive processes. A lecture series at IIIT discussed the *neuroscience of emotion and cognition* specifically. Emotion research remains one of the frontiers of cognitive science.

6.4 Consciousness Revisited

- **Consciousness** sits at the top of the cognitive architecture—a “cloud” area still not fully understood
- For decades, consciousness was **banned** from scientific discussion (considered only a spiritual/religious topic)
- In the last 5–10 years, scientific investigation of consciousness has become acceptable
- It remains a major open problem in cognitive science

6.5 Sensation vs. Perception: Top-Down and Bottom-Up

Sensation vs. Perception

- **Sensation** = incoming sensory input being processed (raw data)
- **Perception** = our *interpretation* and *organisation* of that input (making sense of data)

Top-Down vs. Bottom-Up Processing

- **Top-down processing:** Using *prior knowledge*, templates, and prototypes stored in memory to interpret ambiguous stimuli. Example: reading partially occluded text—you fill in the missing letters from your knowledge of English.
- **Bottom-up processing:** The stimulus itself, through the sensory system, drives perception *without* prior knowledge. Example: the Kanizsa illusion—your visual system creates an apparent shape from the arrangement of elements, regardless of your language or cultural knowledge.

Visual Illusions as Evidence

Kanizsa triangle: You perceive a white triangle that is *not explicitly drawn*—your visual system fills in contours from the arrangement of pac-man shapes. This is **bottom-up** (a Russian speaker with no knowledge of English would see the same triangle).

Occluded text: The word “IBC” is partially hidden behind stripes. You can read it because your **top-down** knowledge of English fills in the gaps. A Russian speaker would *not* be able to decode it—demonstrating top-down processing.

Cafe wall illusion: Perfectly straight parallel lines appear sloped/zigzag due to the arrangement of black-and-white tiles. This is **bottom-up**—an artefact of how your visual system processes contrast.

6.6 The Brain as a Predictive Machine

Predictive Processing Hypothesis

The dominant current theory of brain function: the brain is a **probabilistic prediction machine**.

Using **Bayesian inference**:

$$P(\text{posterior}) = \frac{P(\text{likelihood}) \times P(\text{prior})}{P(\text{normalisation})}$$

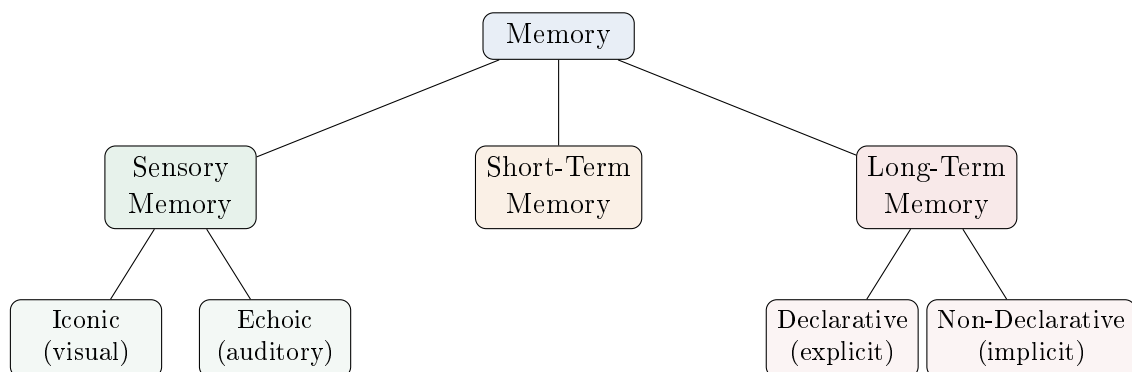
The brain:

1. Maintains a **prior** (what it expects based on past experience)
2. Observes **likelihood** (current sensory evidence)
3. Computes a **posterior** estimate (best guess of current state)
4. Compares prediction with observation; the **difference** = **prediction error**
5. Uses the error to **update the internal model**

This is also called the **predictive coding** hypothesis.

The brain *dislikes ambiguity and uncertainty*—it is always trying to make sense of the world (V.S. Ramachandran).

6.7 Memory Taxonomy (Detailed)



6.7.1 Sensory Memory

- **Duration:** order of seconds (very fleeting)
- **Iconic memory:** visual after-image that persists briefly after stimulus removal
- **Echoic memory:** auditory after-echo (e.g., someone rapidly says a phone number → you can “hear” the echo for a few seconds)
- Erased quickly when new stimuli arrive

6.7.2 Short-Term Memory

- **Duration:** order of minutes to hours
- Example: what you had for breakfast today—you remember it 3 hours later, but likely not tomorrow
- Short-term memories are overwritten by subsequent experiences (lunch erases memory of breakfast)
- Capacity limited (relates to Miller's 7 ± 2)

6.7.3 Long-Term Memory: Declarative (Explicit)

Declarative Memory

Declarative (explicit) memory is memory that can be *consciously recalled and stated*. It has lexical/structural form—you can *explain* what you know.

Two subtypes:

1. **Semantic memory:** knowledge of facts, meanings, and concepts
 - What does “echoic” mean? → Related to “echo” + suffix
 - Who was the first President of India? → Rajendra Prasad
 - Knowledge from textbooks, definitions, general world knowledge
2. **Episodic memory:** vivid recollection of specific personal episodes
 - The day you got your IIIT admission: what you were wearing, who told you, where you were, what time it was
 - Emotionally significant events stored with rich contextual detail

Properties of Declarative Memory

- **Easy to acquire:** repeat something 10 times → memorised (mnemonics help: “OCEAN” for personality types, “All Silver Tea Cups” for trigonometry signs)
- **Easy to lose:** if not rehearsed, declarative memories fade
- Can be *explained* and *communicated* to others

6.7.4 Long-Term Memory: Non-Declarative (Implicit)

Non-Declarative Memory

Non-declarative (implicit) memory cannot be easily verbalised or explained. You *know how* to do something but cannot articulate the precise details.

Two major subtypes:

1. **Procedural (motor) memory:** skills, habits, routines
2. **Priming:** pre-potential of the perceptual system

Procedural Memory.

Properties of Procedural Memory

- **Hard to acquire:** riding a bicycle takes a whole month; driving takes ~2 months
- **Hard to erase:** once learned, you can ride a bicycle years later with no practice (“muscle memory”)
- **Non-declarative:** you cannot fully explain *how* you ride (what angle to lean, what force to apply)
- **Transferable:** write your name with your foot in sand even without practice—the motor *program* transfers across effectors
- **Survives neurodegeneration:** Alzheimer’s patients who lose declarative memory can still ride a bicycle and do mirror-tracing

Brain areas involved: Motor cortex (central sulcus area), cerebellum, basal ganglia. The field of motor memory acquisition is still not fully understood.

Priming.

Priming

Priming is the **pre-potential** of the perceptual/cognitive system so that a future stimulus is recognised more quickly. It operates *below conscious awareness* (subliminally).

- Example: Seeing a bearded man’s face → later, when shown an ambiguous image, you immediately see a bearded figure (visual priming)
- Example: Being primed with wedding-related objects → interpreting a neutral story as being about a wedding
- Advertisers heavily exploit priming (subliminal marketing)

- Priming can be visual, auditory, or cross-modal

6.8 The Motor Homunculus

Homunculus

The **homunculus** (Latin: “little man”) is the **topographic map** of the body on the motor and somatosensory cortex. The body parts are represented in proportion to the *amount of cortical area* devoted to their control, **not** in proportion to their physical size.

Key Features of the Homunculus

The representation is **non-linear**:

- **Face** has a disproportionately *large* cortical area (~ 200 facial muscles $\rightarrow$ enables expressive dance, acting, micro-expressions)
- **Hands** have a large cortical area (fine motor control; **opposable thumb**—a key human adaptation separating us from other animals; enables fine grip)
- **Legs** have a relatively *small* cortical area despite being physically large
- Shown on a **coronal section** at the level of the central sulcus
- The same topographic organisation exists for both **motor** and **somatosensory** cortex

Practice & Review

Q1. [Multiple Choice] Non-declarative (implicit) long-term memory includes:

- (A) Semantic and episodic memory (B) Sensory and working memory
(C) Procedural (motor skill) memory and priming (D) Short-term memory and iconic memory

Answer: C. Non-declarative memory cannot easily be verbalised. It includes procedural memory (“knowing how”—riding a bicycle) and priming (subconscious pre-potential of recognition). Semantic and episodic memories are subtypes of *declarative* (explicit) long-term memory.

Q2. [Short Answer] Outline the complete memory taxonomy, including all types and subtypes with approximate durations.

Answer: (1) **Sensory memory** ($\sim$ seconds): iconic (visual after-image) and echoic (auditory echo); erased by new input. (2) **Short-term memory** ($\sim$ minutes to hours): capacity 7 ± 2 chunks; overwritten by subsequent experiences. (3) **Long-term memory** (days to lifetime): **Declarative (explicit)**—semantic (facts, definitions) and episodic (personal vivid memories with context); **Non-declarative (implicit)**—procedural (motor skills, hard to acquire, hard to erase) and priming (subconscious pre-activation of perception).

- Q3. [Essay]** Explain the Bayesian brain / predictive processing hypothesis. Write the formula, define each term, and describe the prediction error update cycle.

Answer: The brain is a **probabilistic prediction machine**. Formula:

$$P(\text{posterior}) \propto P(\text{likelihood}) \times P(\text{prior})$$

Prior: the brain's expectation based on past experience. **Likelihood:** the current sensory evidence. **Posterior:** the updated best-guess interpretation after combining prior + evidence. **Prediction error:** the mismatch between the brain's prediction and the actual sensory input. The error signal drives *model updating*: the brain revises its internal model to reduce future prediction errors. This is also called **predictive coding**. Implication: what we perceive is largely our brain's prediction—sensory input merely *corrects* the prediction. The brain “dislikes” ambiguity and uncertainty and constantly seeks the most coherent interpretation (Ramachandran).

- Q4. [Multiple Choice]** The motor homunculus shows that the body parts with the *largest* cortical representation are:
 (A) Legs and trunk (B) Shoulders and back
 (C) Face and hands (D) Abdomen and pelvis

Answer: C. Face (~200 facial muscles enable rich expression) and hands (fine motor control; opposable thumb is a key human adaptation) have disproportionately large cortical areas. Representation is proportional to *functional importance*, not physical size.

- Q5. [Short Answer]** Contrast declarative and procedural memory on three dimensions: ease of acquisition, resistance to forgetting, and what happens in Alzheimer's disease.

	Dimension	Declarative	Procedural
<i>Answer:</i>	Acquisition	Easy (repeat ~10 times; mnemonics help)	Hard (weeks–months of practice)
	Forgetting	Easy (fades without rehearsal)	Very difficult (“muscle memory” persists)
	Alzheimer's	<i>Lost early</i> (memory of names, facts)	<i>Preserved</i> (can still ride a bicycle)

- Q6. [Short Answer]** Define priming. Explain the mechanism by which advertisers exploit it.

Answer: Priming is the **pre-potentiation** of the perceptual/cognitive system so that future related stimuli are recognised more rapidly. It operates *below conscious awareness* (subliminally). Advertisers exploit it by repeatedly exposing consumers to brand logos, jingles, or colours in neutral or positive contexts—without the consumer's conscious awareness. Later, encountering the brand activates the pre-potentiated associations (positive feelings, familiarity) and increases likelihood of purchase. Example: a briefly flashed soft-drink logo makes viewers prefer that drink over a competitor's in a subsequent choice.

- Q7. [Essay]** Explain serial vs. parallel processing in the context of human cognition. How does attention function as a resource allocator in a parallel system?

Answer: **Serial processing** (like digital computers): one process at a time; execution is sequential. **Parallel processing** (human brain): multiple simultaneous processes. Evidence: while speaking, a person simultaneously makes hand gestures (motor cortex), maintains visual monitoring (visual cortex), keeps balance (cerebellum), breathes (brainstem)—all running in parallel. A serial processor could not speak and breathe simultaneously. Problem with parallel processing: *information overload*—all channels compete for the same limited processing resources. **Attention** is the solution: it is the resource-allocation mechanism that selectively amplifies relevant processing channels while suppressing irrelevant ones, enabling efficient behaviour in a massively parallel system.

- Q8. [Short Answer]** What makes procedural memory clinically unique? Why does it survive Alzheimer's disease?

Answer: Procedural memory is unique because: (1) **Hard to acquire**—requires weeks/months of practice; (2) **Very hard to lose**—remains after years without practice (“once you know how to ride a bicycle, you never forget”); (3) **Non-verbalizable**—you cannot fully explain the precise motor commands used; (4) **Effector-independent**—write your name with your foot or non-dominant hand; the motor *program* transfers. Alzheimer's primarily destroys the **hippocampus** (episodic, declarative memory). Procedural memory is stored in **motor cortex, cerebellum, and basal ganglia**—structures relatively spared in early Alzheimer's. Hence, patients retain motor skills even as they lose factual and autobiographical memories.

Lecture 7: The Motor System — Control, Reflexes, and Skill Learning (March 27)

7.1 Motor Control: Definitions

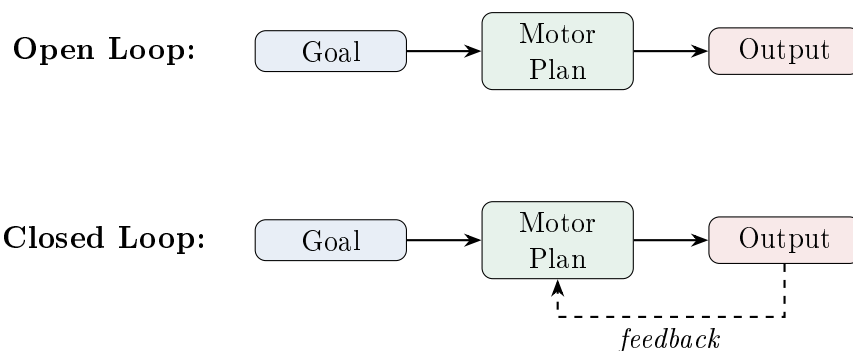
Motor Control

Motor control is the ability to regulate and direct the movements of the body. It is assessed by:

- **Efficiency:** how well a target is hit or a condition is maintained (e.g., bullseye in darts)
- **Robustness/consistency:** how stable performance is across trials (not oscillating wildly)
- **Learning:** repeated practice → measurable improvement in performance scores

The **cerebellum** plays a primary role in motor control and coordination.

7.2 Open-Loop vs. Closed-Loop Control



Open-Loop Control

Open-loop control involves executing a motor action *without* immediate feedback about its outcome. Once initiated, the action runs to completion.

Examples:

- Throwing a football/dart — once released, you cannot adjust mid-flight
- Writing your signature — a practised routine executed without conscious feedback per stroke
- Any well-rehearsed, habitual motor programme

These are typically **memory/habit-driven** actions.

Closed-Loop Control

Closed-loop control involves *continuous feedback* that allows for real-time adjustment of the action.

Examples:

- Driving a car — constantly adjusting speed, direction based on traffic, road conditions
- Threading a needle — perceptual input about needle-eye position guides hand adjustment
- Balancing on an unstable surface

These actions involve a **feedback–adjustment–feedback** cycle.

Many Tasks Are Hybrid

Driving is primarily **closed-loop**, but for an expert on a familiar road, much of it becomes **automatic** (open-loop-like). An unexpected event (construction, rain, oil on road) forces the system back to **closed-loop**. This switch from automatic to deliberate control is an active area of research.

7.3 Handedness

Edinburgh Handedness Inventory

The **Edinburgh Handedness Inventory** is a 100-point questionnaire used by clinicians to determine an individual's **dominant hand**. It asks which hand you use for eating, writing, throwing, cutting, etc.

Handedness and Hemispheric Dominance

- Motor control is **contralateral**: left hemisphere controls right hand and vice versa
- The **left hemisphere is dominant for language** in *most people*—both right-handed *and* left-handed
- Knowing handedness is clinically critical before brain interventions (e.g., surgery for stroke) to identify which hemisphere is more important for the patient
- One evolutionary theory: mothers held infants near the heart (left side) → right hand became free for tool use → right-handedness became dominant
- The world is largely designed for right-handed people (exam desks, scissors, gear shifters, etc.)

7.4 Topographic Organisation and Modularity

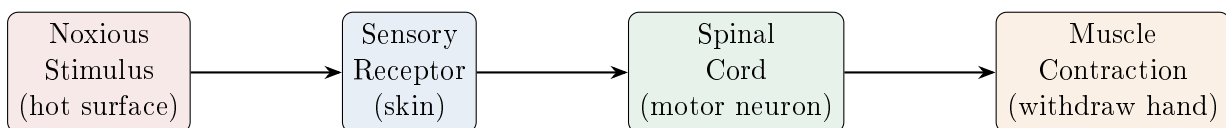
- **Modularity:** different functions (vision, motor, language, auditory) are served by separate brain *modules*
- **Topographic organisation:** within a module, there is a *spatial mapping* of function onto cortical surface (e.g., leg area medial, hand area lateral on motor cortex)
- Not all brain functions are topographically organised, but **sensory and motor functions** are

7.5 Reflexes: The Simplest Motor Circuits

Reflex

A **reflex** is an automatic, involuntary motor response to a sensory stimulus. It does not require conscious processing and is mediated primarily at the **spinal cord** level.

7.5.1 Simple Reflex Arc (Withdrawal Reflex)



Why Reflexes Use the Spinal Cord, Not the Brain

The “long loop” (sensory signal → brain → motor cortex → back down to muscles) would require 4–5 electrochemical hops over a long distance—**too slow** for protective actions. The spinal cord “short circuit” (sensory neuron directly connected to motor neuron at the same spinal level) allows responses within **milliseconds**.

7.5.2 Complex Reflex: Crossed Extensor Reflex

Stepping on a Nail While Walking

If your right foot steps on a nail:

1. **Right leg:** flexor is *excited* (contracts → lifts foot away) + extensor is *inhibited* (relaxes)
2. **Left leg:** extensor is *excited* (stiffens for support) + flexor is *inhibited* (relaxes)

This is the **crossed extensor reflex** (Sherrington, ~1900). It demonstrates that even “simple” reflexes require **bilateral coordination**—muscles always work in **antagonist pairs** (flexor/extensor).

This built-in circuitry has inspired the design of **humanoid robots** and walking machines.

Knee-Jerk Reflex (Patellar Reflex)

A neurologist taps below the kneecap with a rubber hammer. The leg kicks involuntarily.

Purpose: Tests whether the spinal reflex arc is intact. The signal goes from sensory receptor → spinal cord → motor neuron → quadriceps contraction—all without brain involvement.

If absent → possible spinal cord or peripheral nerve damage.

7.6 Hierarchical Motor Control

Motor control operates at **three hierarchical levels**:

Level	Function	Brain Area
Low level (Execution)	Reflexes, spinal circuits, pattern generators	Spinal cord, brainstem
Mid level (Tactical)	Coordination of movement sequences	Motor cortex, cerebellum
High level (Strategic)	Planning; goal-directed behaviour	Association cortex, prefrontal cortex

At each level there are **feedback loops**. Additionally, there is a distinction between:

- **Unconscious control:** most well-learned actions (walking, breathing, maintaining posture)
- **Conscious control:** new or disrupted situations requiring deliberate attention

7.7 Skill Acquisition: From Conscious to Automatic

Stages of Skill Learning

1. **Early stage (high cognitive load):** Every aspect requires conscious attention. Example: a beginner driver must consciously track mirrors, pedals, gear, steering, traffic—this feels like **sensory overload**. Errors are common (e.g., hitting accelerator instead of brake).
2. **Intermediate stage:** Some actions become semi-automatic; conscious resources begin to free up.
3. **Expert stage (low cognitive load):** Most actions are automatic. The control loop “drops back” to lower hierarchical levels. Cognitive resources are freed—an expert driver can hold a conversation while driving.

Key insight: With expertise, the brain shifts from **conscious, cortical control** to **automatic, subcortical/cerebellar control**. This frees up the prefrontal cortex for other tasks. Identifying the brain areas involved in this transition is an active research area.

7.8 Goal-Directed Behaviour

Goal-Directed Behaviour

Goal-directed behaviour involves *planning* and *executing* a sequence of actions to achieve a specific goal. It requires:

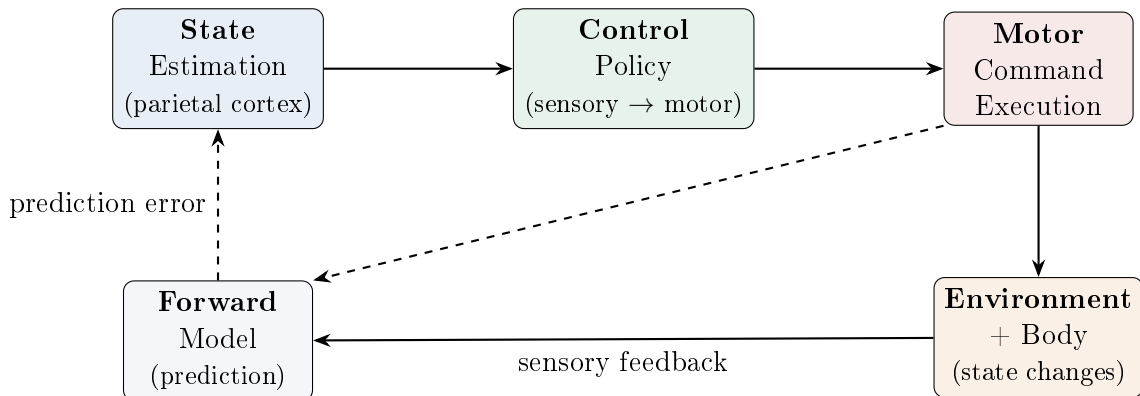
- Decomposing a complex goal into **sub-goals**
- Monitoring progress at each sub-goal
- Adjusting the plan based on feedback

Example: Catching a train from IIIT:

1. Sub-goal 1: Walk/ride to metro station (optimise for time)
2. Sub-goal 2: Ride metro to railway station (optimise: don't stand near door if crowded)
3. Sub-goal 3: Navigate from metro exit to correct platform

Each segment has different optimisation criteria. This requires **conscious, prefrontal** planning.

7.9 The Motor Control Loop: A Control-Systems View



Forward Model

The **forward model** takes a motor command as input and **predicts the expected sensory consequence** of that action *before* sensory feedback arrives. The brain compares this prediction with the actual sensory state; the mismatch is the **prediction error** (or **sensory error**), which drives corrective adjustments.

Inverse Model (Control Policy)

The **inverse model** (or **control policy**) is a mapping from sensory state to appropriate motor action: “Given this sensory state, what motor command should I issue?”

Complete Motor Control Loop

1. **Sense** the current state of the environment (state estimation, parietal cortex)
2. **Decide** on a motor action via the control policy (inverse model)
3. **Execute** the motor command
4. **Predict** the expected sensory consequence (forward model)
5. **Observe** the actual sensory consequence (from environment)
6. **Compute error** = predicted – actual; use error to adjust the policy
7. **Repeat**

Example (driving): You accelerate → forward model predicts you’ll be 1m behind the car in front → actual feedback shows you’re 5m behind (road is slippery) → error signal adjusts future acceleration commands.

7.10 Cost–Reward Evaluation in Motor Planning

- Humans evaluate **costs and rewards** before executing motor commands

- **Degrees of freedom problem:** there are many ways to perform the same action (e.g., picking up a bottle—overhand, underhand, from the side, with detour around obstacles)
- **Energy cost:** the brain optimises for minimal energy expenditure
- **Smoothness:** dancers optimise for flowing, smooth transitions between movements
- **Accuracy:** surgeons optimise for precise, controlled movements
- These cost–reward considerations inform the control policy and are handled by different brain areas

7.11 Brain–Computer Interfaces (BCIs) for Motor Rehabilitation

Building on the BCI concepts from Lecture 4:

- Conditions like **hemiplegia** (one-sided paralysis) leave the motor *intention* system intact but the motor *output* is cut
- BCI systems decode motor intention from brain activity and drive external devices:
 - **Robotic arms / exoskeletons**
 - **Functional electrical stimulation (FES)** of paralysed muscles
- **Surface-based BCI:** non-invasive EEG electrodes on the scalp
- **Implanted BCI:** electrodes implanted inside the brain (e.g., Neuralink by Elon Musk)
- Current research at IIIT: developing a surface-electrode BCI with the neurosurgery department for stroke patients—reading motor intention from healthy-volunteer data → applying decoders to patients for guided physiotherapy

7.12 Deep Brain Stimulation for Parkinson’s Disease (Video Demonstration)

DBS for Parkinson’s: Live Demo

A video demonstration was shown in class:

- Patient has a **resting tremor** (hands and limbs shaking involuntarily)
- An electrode is implanted in the **subthalamic nucleus** (part of basal ganglia)
- When the stimulator is switched **ON**: tremor *dramatically reduces* within seconds; patient can move normally
- When switched **OFF**: tremor gradually returns over minutes
- The patient carries a control switch to toggle stimulation

This intervention is used when Parkinson’s medication is ineffective for advanced cases. It demonstrates the critical role of the **subthalamic nucleus** in motor control and the power of understanding brain circuitry for clinical intervention.

Practice & Review

Q1. [Multiple Choice] Which type of motor control involves continuous feedback and real-time adjustment of an ongoing action?

- (A) Open-loop control (B) Closed-loop control (C) Reflex arc (D) Goal-directed planning

Answer: B. Closed-loop control uses a feedback–adjustment–feedback cycle. Open-loop executes a pre-programmed action without mid-course correction (e.g., throwing a dart—once released, the trajectory is fixed).

Q2. [Short Answer] Describe Sherrington’s crossed extensor reflex. What does it demonstrate about “simple” reflexes?

Answer: When the right foot steps on a nail: **right leg**—flexor muscles excited (contract, lift foot away) *and* extensor muscles inhibited (relax); **left leg**—extensor muscles excited (stiffen to support body weight) *and* flexors inhibited. This demonstrates: (1) reflexes require **bilateral coordination** across both sides of the body; (2) muscles always work in **antagonist pairs** (flexor/extensor); (3) even “simple” spinal reflexes involve complex circuitry. The circuit has inspired the design of humanoid walking robots.

Q3. [Essay] Describe the complete motor control loop, including the concepts of the forward model, inverse model, and prediction error. Use driving on an icy road as your example.

Answer: The loop has six steps: (1) **State estimation** (parietal cortex): sense the current position and velocity. (2) **Inverse model / control policy**: given

the current state, select the appropriate motor command (e.g., how much to accelerate). (3) **Execute** the motor command. (4) **Forward model**: *before* sensory feedback arrives, predict the expected sensory consequence of the action (e.g., “I will be 1m behind the car ahead”). (5) **Observe** actual sensory consequence (e.g., “I am actually 3m behind—the road is more slippery than expected”). (6) **Prediction error** = predicted – actual; this error drives correction of the forward model and the control policy. Driving on ice: the forward model consistently underestimates the slipperiness; prediction errors accumulate and update the model until the driver adapts.

Q4. [Multiple Choice] The Edinburgh Handedness Inventory is a clinical tool used to:

- (A) Diagnose motor neuron disease (B) Measure reaction time accuracy
- (C) Determine an individual’s dominant hand and infer hemispheric dominance
- (D) Assess cerebellar function

Answer: C. The inventory asks which hand is used for activities (writing, throwing, eating). Clinically critical before brain surgery—identifies which hemisphere controls language (typically left) to avoid damaging it. Note: the left hemisphere dominates for language in both right-handed *and* most left-handed individuals.

Q5. [Short Answer] Name the three hierarchical levels of motor control and the brain areas responsible for each.

Answer: (1) **Low level (Execution)**—spinal cord and brainstem: reflex circuits, central pattern generators (rhythmic locomotion); no conscious involvement. (2) **Mid level (Tactical)**—motor cortex and cerebellum: coordination and execution of movement sequences; error correction. (3) **High level (Strategic)**—association cortex and prefrontal cortex: planning, goal decomposition, sub-goal monitoring; requires conscious attention.

Q6. [Essay] How does skill acquisition change the cognitive load and the neural locus of motor control? Why can an expert driver hold a conversation while driving but a novice cannot?

Answer: Skill acquisition progresses through three stages: (1) **Early (high cognitive load)**: all aspects demand conscious, prefrontal attention—mirrors, pedals, gear, traffic, steering simultaneously. Feels like sensory overload; errors are common. (2) **Intermediate**: some actions become semi-automatic; conscious resources begin to free up. (3) **Expert (low cognitive load)**: most actions are fully automatic, executed by subcortical circuits (motor cortex, cerebellum, basal ganglia). The prefrontal cortex is no longer needed for routine control and is *freed* for other tasks—hence an expert can converse while driving. This transition from cortical → subcortical control is an active research area (identifying the brain areas involved in this “handoff”). Implication: automaticity is a sign of expertise, not inattention.

Q7. [Multiple Choice] The subthalamic nucleus targeted by deep brain stimulation (DBS) for Parkinson’s disease is part of the:

(A) Cerebellum (B) Thalamus (C) Basal ganglia (D) Hippocampus

Answer: C. The subthalamic nucleus is part of the basal ganglia (the motor output gateway). DBS of this nucleus delivers continuous mild electrical stimulation, normalising aberrant firing patterns and dramatically reducing Parkinson's tremors and rigidity.

Q8. [Short Answer] Explain the degrees of freedom problem in motor planning. How does the brain resolve it?

Answer: The same goal (e.g., picking up a water bottle) can be achieved through *many* different movement trajectories (overhand grip, underhand grip, from the side, with detour around an obstacle). The motor system has far more degrees of freedom than necessary—it must select *one* specific trajectory. The brain resolves this via **cost–reward evaluation**: minimising energy expenditure, maximising smoothness (dancers), maximising accuracy (surgeons), or optimising for time. Different optimisation criteria engage different brain areas. Research in motor control studies which cost functions the brain implicitly minimises.

Lecture 8: Motor Control Continued — Internal Models, Cost & BCIs (April 7)

This lecture extends the motor system treatment from March 27 by formalising the brain’s *internal models* of itself and the world, introducing cost–reward evaluation, and deepening the treatment of brain–computer interfaces (BCIs).

8.1 Stationary vs. Non-Stationary Systems

Stationary vs. Non-Stationary

A **stationary system** has statistical features (mean, variance, distributions) that remain constant over time.

A **non-stationary system** has parameters that *change continuously* over time. The real world — weather, traffic, body state, social interactions — is overwhelmingly non-stationary.

Why Non-Stationarity Matters for Motor Control

The brain must constantly *re-estimate* the state of the body and the environment because both are changing. Examples:

- Your muscle strength changes with fatigue over a single tennis match
- Road friction changes with rain
- Weight of a grocery bag changes as you add items

A motor control system that assumed a stationary world (fixed parameters) would fail rapidly in the real world. Hence the need for **adaptive internal models** that continuously update.

8.2 Forward and Inverse (“Backward”) Models (Deeper)

Recall the basic control loop from Lecture 7. This lecture formalises the two key internal models:

Forward Model

The **forward model** takes as input a *motor command* and produces a prediction of the *expected sensory consequence* of executing that command.

$$\text{motor command} \xrightarrow{\text{forward model}} \text{predicted sensory outcome}$$

Example: Before you throw a ball, the forward model predicts where your hand will be and how the ball will leave it; this allows you to adjust mid-throw and correct errors.

Inverse Model / Backward Model

The **inverse model** (also called *backward model* or *control policy*) is the reverse mapping: given a *desired sensory state* (goal), it outputs the *motor command* required to achieve it.

$$\text{desired state} \xrightarrow{\text{inverse model}} \text{required motor command}$$

Example: “I want to pick up this cup” (desired state) $\rightarrow$ “contract biceps, rotate wrist, extend fingers” (motor command).

Use of “Backward Model”

In this lecture’s usage, the “backward model” is sometimes used to mean *state estimation*: inferring the current state of the world by working backward from sensory observations. Whether “backward” denotes the inverse controller or state inference depends on context; both run *opposite* to the forward model.

8.3 State Estimation

State Estimation

State estimation is the process of inferring the current state of the body and environment (positions, velocities, forces) from:

1. **Sensory feedback** — proprioception, vision, vestibular, touch
2. **Prior knowledge** — internal models of physics, body mechanics
3. **Efference copy** — a copy of the motor command sent back for prediction

Latent variables (state variables not directly observable) are estimated statistically, similar to latent-variable estimation in machine learning.

Brain area: **Parietal cortex** plays a key role in state estimation — maintains a continuous “map” of where the body is in space.

8.4 Cost Estimation and Reward

Cost–Reward Optimisation in Motor Planning

The brain does not simply execute any valid motor plan; it *optimises* over many possible plans. For each candidate action, the brain estimates:

- **Cost:** energy expenditure, time, risk, pain, attentional resources
- **Reward:** goal completion, satisfaction, survival value

The selected action maximises (reward – cost). This resolves the **degrees-of-freedom problem** (many paths to the same goal; pick the optimal one).

Brain area for cost/reward evaluation: Basal ganglia — evaluates action values and gates motor output accordingly. Dysfunction of basal ganglia (e.g., Parkinson’s) impairs this selection process.

8.5 Brain Areas Involved in Motor Control (Integrated View)

Brain Area	Role in Motor Control
Motor cortex	Generates motor commands
Cerebellum	Coordinates movement; error correction; forward model implementation
Basal ganglia	Cost/reward evaluation; action selection; motor gateway
Parietal cortex	State estimation; spatial representation of body/environment
Prefrontal cortex	Goal-directed planning; strategic motor sequencing
Spinal cord	Reflex arcs; pattern generators; low-level execution

8.6 Skill Learning — Three Stages (Fitts & Posner)

Stages of Skill Acquisition

Learning a new motor skill progresses through three stages:

1. **Cognitive stage** — high cognitive load; learner explicitly thinks through each step; many errors. Example: first driving lesson (checking mirrors, pedals, gear deliberately).
2. **Associative stage** — refinement; errors decrease; components start linking together. Example: driving feels more coordinated but still demands attention.
3. **Automatic stage** — skill executes with minimal conscious effort; cognitive resources freed. Example: expert driver converses while driving.

As expertise develops, the locus of control *shifts* from cortical (prefrontal, motor cortex — cognitive stage) to subcortical (basal ganglia, cerebellum — automatic stage).

8.7 Brain–Computer Interfaces (Expanded)

Brain–Computer Interface (BCI) / Brain–Machine Interface (BMI)

A **BCI** (or **BMI**) is a system that decodes brain signals and uses them to control external devices — computers, prosthetic limbs, exoskeletons, communication aids — *bypassing* normal motor output pathways.

Two major categories:

- **Non-invasive BCI:** EEG electrodes on the scalp. Safer but noisier signal. Used in stroke rehabilitation and research.
- **Invasive BCI:** Electrodes placed directly on the cortex (ECoG) or implanted into tissue (e.g., Utah arrays, Neuralink). Much higher signal quality; enables complex control (speech decoding, cursor control, robotic arm manipulation).

BCI Applications

1. **Prosthetic limbs** — amputees control prosthetic hands/arms via decoded motor intent
2. **Exoskeletons** — wearable robotic skeletons controlled by brain signals; restore walking in paralysed patients
3. **Communication devices** — for ALS, locked-in syndrome (Stephen Hawking used a muscle-based interface; newer BCIs decode intended speech directly)
4. **Functional Electrical Stimulation (FES)** — BCI decodes intent, FES stimulates paralysed muscles
5. **Enhancement applications** — search-and-rescue, military exoskeletons (ethically debated)

Neural Plasticity and BCIs

A remarkable finding: when BCI users practise with a prosthetic, the brain gradually treats the external device *as if it were part of the body*. Motor maps reorganise to incorporate the prosthetic. This demonstrates the extraordinary **adaptability** of the brain — it can extend its body schema to include artificial effectors.

8.8 Neural Prosthetics

Neural Prosthetic

A **neural prosthetic** is a device that replaces or restores a neural function. Examples:

- **Cochlear implant** — restores hearing by electrically stimulating auditory nerve
- **Retinal prosthetic** — restores vision by stimulating optic nerve/retina
- **Motor prosthetic** — decodes motor intent from cortex to drive external actuator
- **Deep Brain Stimulator** — modulates activity (Parkinson's, OCD)

Practice & Review

Q1. [Multiple Choice] A non-stationary system is characterised by:

- (A) Constant parameters over time (B) Parameters that change continuously over time
 (C) No feedback loop (D) A single degree of freedom

Answer: B. Non-stationary systems have time-varying statistical features (mean, variance). The real world is overwhelmingly non-stationary; the brain must continuously re-estimate state.

- Q2. [Short Answer]** Distinguish between the forward model and the inverse model in motor control. What does each compute?

Answer: The **forward model** maps *motor command* $\rightarrow$ *predicted sensory consequence*; it predicts “what will happen if I execute this command.” The **inverse model** (control policy) maps *desired state* $\rightarrow$ *motor command required*; it answers “what command do I need to achieve this goal?” Both are essential: the inverse model plans; the forward model predicts outcomes for error correction.

- Q3. [Essay]** Explain the degrees-of-freedom problem in motor control. How does the brain resolve it, and which brain area is centrally involved?

Answer: The degrees-of-freedom problem: there are many possible ways to execute the same movement (e.g., picking up a bottle — overhand, underhand, via detour). The motor system has redundancy, yet must pick a single trajectory. The brain resolves this through **cost–reward optimisation**: for each candidate action, it estimates costs (energy, time, risk) and expected rewards (goal achievement, satisfaction), then selects the action that maximises (reward – cost). Different criteria dominate in different contexts: dancers optimise smoothness, surgeons optimise precision, tired workers optimise energy. The **basal ganglia** are central to this selection process; dysfunction (e.g., Parkinson’s) impairs action selection.

- Q4. [Multiple Choice]** Which brain area is most central to *cost and reward evaluation* of motor actions?

(A) Motor cortex (B) Cerebellum (C) Basal ganglia (D) Brainstem

Answer: **C.** The basal ganglia evaluate action values and gate motor output; they are the key substrate for action selection under competing alternatives.

- Q5. [Short Answer]** What is state estimation? Name two sources of information the brain uses for it.

Answer: State estimation is the inference of the current state of the body and environment (positions, velocities, forces of body parts and external objects). Sources: (1) **sensory feedback** (proprioception, vision, vestibular, touch); (2) **prior knowledge** (internal models of physics and body mechanics); (3) **efference copy** (copy of the motor command). The **parietal cortex** is the primary area for state estimation.

- Q6. [Essay]** Describe the three stages of skill learning (Fitts & Posner). How does the neural locus of control shift across stages?

Answer: (1) **Cognitive stage**: high cognitive load; every component requires conscious attention; many errors (novice driver consciously tracking mirrors, pedals, gears). (2) **Associative stage**: components start chaining; errors decrease; performance more fluid but still demanding. (3) **Automatic stage**: minimal conscious effort; cognitive resources freed (expert driver converses while driving). Neural shift: control transfers from **cortical** structures (prefrontal, motor cortex — effortful, deliberate) to **subcortical** structures (basal ganglia, cerebellum — rapid, automatic). This transfer frees the prefrontal cortex for higher goals.

Q7. [Multiple Choice] Which of the following is *not* a typical use of brain–computer interfaces (BCIs)?

- (A) Controlling a prosthetic limb via decoded motor intent
- (B) Helping ALS patients communicate
- (C) Replacing the hippocampus entirely
- (D) Driving an exoskeleton for stroke rehabilitation

Answer: C. Current BCIs decode neural signals to control external devices; they do not replace complex memory-forming structures like the hippocampus. Prosthetic control, communication, and rehabilitation are all active applications.

Q8. [Short Answer] Contrast invasive and non-invasive BCIs. Give one example of each.

Answer: Non-invasive BCIs use EEG electrodes on the scalp. They are safe and require no surgery, but have low signal quality (signal attenuated by skull/scalp). Example: EEG-based wheelchair controller for stroke patients. **Invasive BCIs** place electrodes directly on or in the cortex (ECoG, Utah arrays, Neuralink). They give much higher signal quality and enable complex control (speech decoding, high-dimensional robotic control) but require brain surgery and carry infection risk. Example: Utah-array implant enabling paralysed patient to type via cursor control or control a robotic arm.

Q9. [Essay] Why is the brain’s ability to incorporate a prosthetic into its body schema a remarkable finding? What does it suggest about neural plasticity?

Answer: With practice, BCI users’ motor maps reorganise to treat the external device *as if it were part of their own body*. This is profound because: (1) it shows the body schema is not hard-coded but plastic; (2) the brain can extend its motor and sensory representations to include artificial effectors; (3) this plasticity enables long-term mastery of prosthetics as natural extensions rather than tools to be consciously operated. It also opens avenues for augmentation (hybrid cyborg systems) and raises ethical questions about enhancement.

Q10. [Short Answer] Name four brain areas involved in motor control and give the role of each.

Answer: (1) **Motor cortex** — generates motor commands; (2) **Cerebellum** — coordinates movements, implements forward models, corrects errors; (3) **Basal ganglia** — action selection via cost/reward; motor gateway; (4) **Parietal cortex** — state estimation, spatial body representation; (5) **Prefrontal cortex** — goal-directed planning; (6) **Spinal cord** — reflexes and low-level execution.

Lecture 9: Intelligence and Learning (April 10)

Supplementary PDF

A handwritten/slide PDF titled `IBC_Intelligence_10Apr2026.pdf` accompanies this lecture in the `April-10/` folder. Refer to it for additional visual summaries of Gardner's and Sternberg's theories.

9.1 Reinforcement Learning — Bridge from Motor Control

Reinforcement Learning

Reinforcement learning is a form of learning in which behaviour is modified by its *consequences* (rewards or punishments). Over many trials, actions that lead to rewards are strengthened, and actions that lead to punishments are weakened.

Model-Based vs. Model-Free Learning

- **Model-based learning:** the agent builds an internal *model* of the world's dynamics (state transitions, rewards) and uses it to *plan* optimal actions. Flexible but computationally expensive.
- **Model-free learning:** the agent directly learns action values by *trial and error*, without an explicit world model. Faster and simpler but less flexible; can fail when the environment changes.

Humans (and AI) use both. The same terminology extends to machine-learning algorithms (Q-learning = model-free; MCTS/Dyna = model-based).

Neural correspondence: As skills transition from cognitive to automatic (Lecture 8), brain activity moves between areas—often from prefrontal (model-based planning) to striatal/cerebellar (model-free habits).

9.2 Intelligence — Definitions

Intelligence

Intelligence is the ability to *learn, understand, reason, and adapt* to new situations. It encompasses multiple cognitive capacities: verbal comprehension, spatial reasoning, memory, problem-solving, and more. Intelligence is studied in *psychology* (a different field from neuroscience per se).

9.3 Theories of Intelligence

9.3.1 Spearman's Two-Factor Theory (1904)

Spearman's g Factor

Charles Spearman proposed that all cognitive abilities share a **general intelligence factor** (g) plus task-specific factors (s).

- Evidence: performance on different tests is *positively correlated* (a person who does well on verbal tests often does well on spatial tests too)
- This “positive manifold” implies a common underlying capacity (g)
- g is often identified with IQ; remains influential today

9.3.2 Thurstone's Primary Mental Abilities (1938)

Thurstone's 7 Primary Abilities

Louis Thurstone rejected a single g factor and proposed **seven distinct primary mental abilities**: (1) verbal comprehension; (2) word fluency; (3) number (arithmetic); (4) spatial visualisation; (5) associative memory; (6) perceptual speed; (7) reasoning

9.3.3 Gardner's Multiple Intelligences (1983)

Gardner's Theory of Multiple Intelligences

Howard Gardner proposed at least **8 independent intelligences**, each reflecting a distinct ability:

1. **Linguistic** — word-smart (writers, poets)
2. **Logical-mathematical** — number-smart (mathematicians, scientists)
3. **Spatial** — picture-smart (architects, sculptors)
4. **Bodily-kinesthetic** — body-smart (athletes, dancers, surgeons)
5. **Musical** — music-smart (composers, performers)
6. **Interpersonal** — people-smart (leaders, counsellors)
7. **Intrapersonal** — self-smart (self-awareness, introspection)
8. **Naturalistic** — nature-smart (biologists, farmers)

Implication: a child who struggles with maths but excels in music is not “less intelligent”—they exhibit a different profile of intelligences.

9.3.4 Sternberg's Triarchic Theory

Sternberg's Three Intelligences

Robert Sternberg proposed intelligence has three components:

1. **Analytical** — problem-solving, logical reasoning (what IQ tests measure)
2. **Creative** — novel situations, innovation, imagination
3. **Practical** — “street smarts”; real-world problem-solving; adapting to everyday contexts

9.3.5 Guilford's Structure of Intellect

J.P. Guilford proposed intelligence is the product of *operations* (cognition, memory, evaluation, etc.) $\times$ *contents* (visual, auditory, symbolic, etc.) $\times$ *products* (units, classes, relations, etc.). This combinatorial model yields over 120 distinct intellectual abilities.

9.4 Fluid vs. Crystallised Intelligence (Cattell)

Fluid and Crystallised Intelligence

Raymond Cattell distinguished two broad types:

- **Fluid intelligence (Gf)** — ability to reason, think abstractly, and solve *novel* problems without reliance on past knowledge. Peaks in young adulthood; *declines with age*.
- **Crystallised intelligence (Gc)** — accumulated knowledge, vocabulary, and expertise. *Stable or improves* with age (older adults know more; they know it more deeply).

Why Cognitive Decline Is Not Total

Older adults lose fluid intelligence (slower reasoning) but retain crystallised intelligence (knowledge). Experienced professionals often compensate for declining fluid abilities with extensive domain knowledge.

9.5 Measuring Intelligence — IQ Tests

Major IQ Tests

- **Binet–Simon Scale (1905)** — *Alfred Binet* designed the first practical intelligence test in France to identify children needing educational support
- **Stanford–Binet Intelligence Scale** — American adaptation (Lewis Terman, Stanford University)
- **Wechsler Adult Intelligence Scale (WAIS)** — standard adult IQ test (measures verbal IQ, performance IQ, full-scale IQ)
- **Wechsler Intelligence Scale for Children (WISC)** — child version

Classical IQ formula:

$$\text{IQ} = \frac{\text{Mental Age}}{\text{Chronological Age}} \times 100$$

A child whose mental age matches their chronological age has IQ = 100. Modern IQ scoring uses deviation scores normed to a population mean of 100 and SD of 15.

Limitations and Criticisms of IQ

IQ scores are *not* absolute measures of intelligence:

- Cultural bias in test items
- Test-taking anxiety and environment affect scores
- Only samples a narrow slice of abilities (missing creativity, emotional, practical intelligence)
- Risk of misuse for educational sorting, employment decisions, and social policy (see criticisms of *The Bell Curve*)

9.6 Emotional Intelligence (EI / EQ)

Emotional Intelligence

Emotional intelligence (EI) is the ability to *recognise* and *understand* emotions in oneself and others, and to use this awareness to *guide thought and behaviour*. Popularised by **Daniel Goleman**'s book *Emotional Intelligence* (1995).

Five Components of Emotional Intelligence (Goleman)

1. **Self-awareness** — knowing your own emotions
2. **Self-regulation** — managing your emotions (impulse control)
3. **Motivation** — intrinsic drive toward goals
4. **Empathy** — recognising and understanding others' emotions
5. **Social skills** — navigating relationships effectively

EI vs. IQ: High IQ does not predict high EI, and vice versa. EI is a separate (though related) construct. EI is especially important for leadership, teamwork, and mental well-being.

9.7 Developmental Psychology and Intelligence

Cognitive Development Milestones

- Different cognitive abilities develop at *different rates* during childhood
- **Working memory** capacity increases significantly between ages 5 and 9
- **Frontal lobe** (executive function, impulse control) continues to mature until ~21
- Early childhood is a “sensitive period” for language acquisition

Understanding developmental milestones is crucial for curriculum design and educational psychology.

9.8 Cognitive Decline and Aging

- **Normal aging** brings gradual decline in processing speed, working memory, fluid intelligence
- **Crystallised intelligence** (vocabulary, knowledge) remains stable or improves
- Influenced by genetics, lifestyle (exercise, diet, social engagement), and education
- **Assessment tools:**
 - **MMSE** (Mini-Mental State Examination) — brief cognitive screening
 - **MoCA** (Montreal Cognitive Assessment) — more sensitive screening

9.9 Nature vs. Nurture in Intelligence

Nature–Nurture in IQ

Twin studies and adoption studies show intelligence is influenced by *both*:

- **Nature** (genes) — identical twins raised apart show high IQ correlations (~ 0.7)
- **Nurture** (environment) — adopted children’s IQs correlate with adoptive family environment; enrichment and education raise IQ scores

Heritability estimates range 0.5–0.8 depending on methodology and population. Both contribute; neither is solely responsible.

9.10 Giftedness

Giftedness

Giftedness refers to exceptional abilities or talents that substantially exceed typical peer levels. Gifted individuals may excel in academics, arts, athletics, or specific domains.

- **Identification:** IQ tests, achievement tests, domain-specific evaluations
- **Education:** gifted children often need enriched or accelerated curricula to avoid boredom and disengagement
- **Early intervention** (either for gifted or lagging children) is critical for realising potential

Practice & Review

Q1. [Multiple Choice] Which theory proposes that all cognitive abilities share a general intelligence factor, g ?

- (A) Gardner’s multiple intelligences (B) Spearman’s two-factor theory
(C) Sternberg’s triarchic theory (D) Thurstone’s primary mental abilities

Answer: B. Spearman proposed that g underlies all specific abilities. Evidence: positive correlations among diverse cognitive tests (the “positive manifold”).

Q2. [Short Answer] Distinguish between model-based and model-free reinforcement learning.

Answer: Model-based learning: the agent builds an internal model of the world’s dynamics (states, transitions, rewards) and uses it to *plan*. Flexible but slow. **Model-free learning:** the agent directly learns action values by *trial and error* without an explicit model. Fast and simple but inflexible when the environment changes. Humans use both; the balance shifts toward model-free habits as skills become automatic.

- Q3. [Essay]** Compare Gardner's multiple intelligences theory with Spearman's *g* factor. What are the practical implications of each for education?

Answer: Spearman proposes a *single general factor (g)* plus task-specific factors. Evidence: positive manifold (correlated test performance). Gardner rejects a single *g* and instead proposes *eight independent intelligences* (linguistic, logical-mathematical, spatial, bodily-kinesthetic, musical, interpersonal, intrapersonal, naturalistic). **Educational implications:** under Spearman, children are ranked on a single IQ scale and curricula target *g*. Under Gardner, education should recognise diverse talent profiles: a child low on logical-mathematical but high on musical intelligence needs different teaching methods, not remediation. Gardner's view motivates arts and vocational education alongside academics; Spearman's view supports standardised testing and sorting by general ability.

- Q4. [Multiple Choice]** Which theory proposes that intelligence comprises analytical, creative, and practical components?

(A) Triarchic Theory (Sternberg) (B) Multiple Intelligences (Gardner)
(C) Structure of Intellect (Guilford) (D) Two-Factor Theory (Spearman)

Answer: **A.** Sternberg's triarchic theory: analytical (academic problem-solving), creative (novel situations), practical (street smarts/everyday adaptation).

- Q5. [Short Answer]** What is the difference between fluid intelligence (*Gf*) and crystallised intelligence (*Gc*)? How does each change with age?

Answer: **Fluid intelligence:** ability to reason, think abstractly, and solve *novel* problems without prior knowledge. *Declines with age* (slower processing, reduced working memory). **Crystallised intelligence:** accumulated knowledge, vocabulary, and expertise. *Stable or improves with age*. Older adults often compensate for fluid decline with richer crystallised stores (e.g., an experienced doctor recognises patterns quickly from years of cases even though raw reasoning speed has slowed).

- Q6. [Essay]** Explain emotional intelligence, its five components (Goleman), and why high IQ does not guarantee high EI.

Answer: **Emotional intelligence (EI):** ability to recognise and understand emotions in self and others, and use this awareness to guide thought and behaviour (popularised by Daniel Goleman in 1995). Five components: (1) **self-awareness** (knowing your emotions); (2) **self-regulation** (managing impulses); (3) **motivation** (intrinsic drive); (4) **empathy** (recognising others' emotions); (5) **social skills** (relationship navigation). Why high IQ $\neq$ high EI: IQ measures analytical/logical reasoning; EI measures a distinct set of social-emotional skills. A brilliant scientist with poor empathy may struggle as a team leader; a leader with moderate IQ but excellent EI can inspire teams. Both contribute to success; they are orthogonal constructs.

- Q7. [Multiple Choice]** Who developed the first practical intelligence test, designed to identify children needing special educational support?

(A) Lewis Terman (B) David Wechsler (C) Alfred Binet (D) Francis Galton

Answer: **C**. Alfred Binet, with Theodore Simon, developed the Binet–Simon Scale (1905) in France. Terman adapted it as the Stanford–Binet. Wechsler later created the WAIS and WISC.

Q8. [Short Answer] Write the classical formula for IQ and interpret an IQ score of 100.

Answer: Classical formula: $IQ = (\text{Mental Age} / \text{Chronological Age}) \times 100$. An IQ of 100 means the child’s mental age equals their chronological age — they perform at the average level for their age. Modern IQ scoring uses deviation scores normed to mean 100, SD 15. Scores above 130 are often called “gifted”; scores below 70 suggest significant cognitive impairment.

Q9. [Essay] Discuss the nature–nurture debate in the context of intelligence. What do twin and adoption studies tell us?

Answer: The debate: is intelligence primarily genetic (nature) or environmental (nurture)? Both contribute. Evidence: **Identical twins raised apart** show high IQ correlations (~ 0.7), higher than unrelated individuals raised together — suggests strong genetic component. **Adopted children** show IQs correlating with both biological parents (genetic) and adoptive environment (enrichment). Heritability estimates are 0.5–0.8. However, *heritability* describes variance within a population, not destiny for individuals. Environment (nutrition, education, stimulation, stress) strongly modulates IQ; the Flynn effect (rising IQs across generations) shows environmental influence at population scale. Both factors interact; nurture can substantially raise or lower realisation of genetic potential.

Q10. [Short Answer] What is “giftedness”? Why is early intervention important for both gifted and lagging children?

Answer: **Giftedness:** exceptional abilities exceeding peers (academic, artistic, athletic, etc.). **Early intervention matters** because: (1) gifted children who are not challenged may become bored, disengaged, or develop behavioural problems; accelerated/enriched curricula help realise potential; (2) lagging children benefit most from targeted support in sensitive developmental windows (early childhood, when working memory, language, and executive function are rapidly forming); (3) both groups risk being overlooked by one-size-fits-all curricula. Early identification via assessments (IQ tests, achievement tests, behavioural observations) allows tailored educational plans.

Lecture 10: Memory Part 1 — Formation, Stores, and Models (April 14)

Supplementary PDF

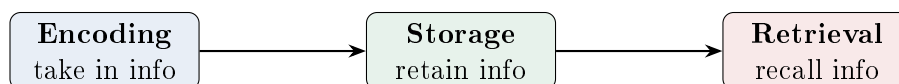
A handwritten/slide PDF titled **Memory IBC 2026 part 1.pdf** accompanies this lecture in the **April-14/** folder. Refer to it for diagrams of the Atkinson-Shiffrin and Baddeley models.

10.1 What is Memory?

Memory

Memory is the brain's ability to *encode*, *store*, and *retrieve* information. It is essential for learning, adaptation, identity, and survival. Without memory, every experience is novel; no learning accrues.

10.2 Three Stages of Memory



The Three Memory Stages

1. **Encoding** — converting sensory input into a form that can be stored. Requires *attention*; what you don't attend to, you won't remember.
2. **Storage** — maintaining encoded information over time. Can be short-term (seconds to minutes) or long-term (hours to lifetime).
3. **Retrieval** — accessing stored information when needed. Can be *free recall* (no cues), *cued recall* (with prompts), or *recognition* (familiarity judgement).

Failures at any stage manifest as “forgetting,” but the underlying cause may differ (poor encoding vs. decay vs. retrieval failure).

10.3 Types of Memory (Detailed)

Lecture 6 introduced the memory taxonomy. This lecture drills into the *temporal* and *structural* aspects.

10.3.1 Sensory Memory

- **Duration:** very brief — milliseconds to a few seconds
- **Capacity:** large (everything the senses detect, briefly)

- **Iconic memory** — visual after-image (e.g., spinning sparkler trail)
- **Echoic memory** — auditory after-echo (“wait, what did you say?” and you can hear it again briefly)
- Information decays rapidly unless attended

10.3.2 Short-Term Memory (STM)

- **Duration:** ~15–30 seconds (without rehearsal)
- **Capacity:** limited to 7 ± 2 chunks (Miller’s law, revisited)
- Overwrites with new information if not rehearsed
- Example: remembering a phone number long enough to dial it

10.3.3 Long-Term Memory (LTM)

- **Duration:** hours to a lifetime
- **Capacity:** effectively unlimited
- Requires *consolidation* from STM
- Subtypes: **declarative** (semantic + episodic) and **non-declarative** (procedural + priming) — see Lecture 6

10.3.4 Working Memory

Working Memory (Baddeley)

Working memory is not just passive short-term storage — it is the active *holding and manipulation* of information for ongoing cognitive tasks (reasoning, comprehension, mental arithmetic). Proposed by **Alan Baddeley** as distinct from (and more elaborate than) simple STM.

10.4 Atkinson–Shiffrin Multi-Store Model (1968)

The Modal Model of Memory

The classical **Atkinson–Shiffrin** model describes three sequential stores:

Sensory Memory $\xrightarrow{\text{attention}}$ Short-Term Memory $\xrightarrow{\text{rehearsal}}$ Long-Term Memory

- **Attention** gates transfer from sensory $\rightarrow$ STM
- **Rehearsal** (repetition) transfers STM $\rightarrow$ LTM
- **Retrieval** pulls LTM content back into STM for use
- Information in STM that is not rehearsed decays and is lost

Simple but influential; later models (Baddeley) refine the STM component.

10.5 Baddeley’s Working Memory Model

Baddeley’s Four-Component Model

Alan Baddeley proposed that working memory has multiple specialised components:

1. **Central Executive** — attentional control system; directs focus, switches tasks; resides in *prefrontal cortex*
2. **Phonological Loop** — stores and rehearses *verbal/acoustic* information (inner voice; repeating a phone number)
3. **Visuospatial Sketchpad** — stores and manipulates *visual and spatial* information (mentally rotating an object)
4. **Episodic Buffer** (added later, 2000) — integrates information across modalities into unified episodes; links working memory with long-term memory

Phonological Loop in Action

When you learn a new foreign vocabulary word, you typically repeat it to yourself multiple times. That silent rehearsal is the *phonological loop* at work. Dual-task experiments show that loading the phonological loop with irrelevant speech (“the, the, the...”) impairs verbal learning but leaves visuospatial task performance intact — evidence for separate modality-specific components.

10.6 Brain Regions for Memory

Brain Region	Memory Role
Hippocampus	Forms new declarative memories; consolidates STM → LTM; crucial for episodic memory and spatial navigation
Amygdala	Emotional memories; enhances consolidation of emotionally significant events; fear conditioning
Prefrontal Cortex	Working memory (central executive); attentional control; retrieval and manipulation of stored memories; source monitoring
Medial Temporal Lobe	Familiarity, recognition (parahippocampal, perirhinal cortex around hippocampus)
Cerebellum and Basal Ganglia	Procedural (motor) memory; habits
Neocortex (distributed)	Long-term storage of consolidated memories

10.7 Amnesia — Anterograde and Retrograde

Types of Amnesia

- **Anterograde amnesia:** inability to form *new* memories after an injury/event. Old memories (before injury) are preserved.
- **Retrograde amnesia:** loss of memories from *before* an injury. Ability to form new memories may be intact.

Both types result from damage to the hippocampus and surrounding medial temporal structures.

Patient H.M. (Henry Molaison)

H.M. had his hippocampus and nearby medial temporal lobe tissue surgically removed (1953) to treat severe epilepsy. The surgery cured his seizures but left him with profound **anterograde amnesia** — he could not form new declarative memories for the rest of his life (every daily experience was instantly forgotten). However:

- His *old* memories (pre-1953) were largely intact
- His *working memory* was normal (could hold a phone number)
- His *procedural memory* was intact (he could learn new motor skills, though without remembering having learned them)

H.M.'s case proved that the hippocampus is essential for *forming new declarative memories* but not for procedural or short-term memory — demonstrating memory systems are *modular* and separable.

10.8 Consolidation

Memory Consolidation

Consolidation is the process by which short-term memories are stabilised and transferred to long-term storage. It involves:

- **Synaptic consolidation** (hours) — protein synthesis and synaptic strengthening
- **Systems consolidation** (days to years) — gradual transfer of hippocampus-dependent memories to distributed neocortical storage; memories become less reliant on the hippocampus over time

10.9 Factors Influencing Memory

What Makes Memories Stick

1. **Attention** — no attention $\Rightarrow$ no encoding (the most common cause of “forgetting” is never encoding in the first place)
2. **Emotion** — emotionally charged events are consolidated more vividly (amygdala-hippocampus interaction)
3. **Context** — physical setting, mood, smells provide retrieval cues (*context-dependent memory*)
4. **State** — physiological/emotional state at encoding influences retrieval (*state-dependent learning*: studying stressed helps recall when stressed)
5. **Elaboration** — deeper semantic processing outperforms shallow processing
6. **Repetition** — especially *spaced repetition* (reviewing at increasing intervals)
7. **Sleep** — critical for consolidation; details in Lecture 11

10.10 Levels of Processing (Craik & Lockhart)

Shallow vs. Deep Processing

- **Shallow processing** — surface features (font, sound); e.g., “Is the word in capitals?”
- **Intermediate** — acoustic properties; e.g., “Does it rhyme with *cat*?”
- **Deep processing** — semantic/meaning-based; e.g., “Does this word fit in this sentence?”

Key finding: deeper processing $\Rightarrow$ better retention. Semantic encoding creates richer associations in memory. Useful study tip: ask *why* and *how* a fact matters, not just memorise its surface form.

10.11 Memory Strategies (Mnemonics)

- **Chunking** (Miller) — group items into larger meaningful units
- **Acronyms** — e.g., OCEAN for the Big Five personality traits
- **Method of loci** — place items mentally along a familiar walk
- **Rhymes and songs** — e.g., “Thirty days hath September...”
- **Visual imagery** — create vivid images linking items
- **Elaborative encoding** — connect new info to existing knowledge

- **Spaced repetition** — review at expanding intervals (flashcard apps implement this)

Practice & Review

Q1. [Multiple Choice] The primary role of the hippocampus in memory is to:

- (A) Store emotional memories permanently (B) Consolidate new information from short-term to long-term memory
(C) Execute motor habits (D) Regulate heart rate during stress

Answer: B. The hippocampus is essential for forming new declarative memories and consolidating STM → LTM. Damage causes *anterograde amnesia* (inability to form new memories). The amygdala handles the *emotional tagging*; storage itself is distributed across the neocortex after consolidation.

Q2. [Short Answer] What is working memory? How is it related to (and distinct from) short-term memory?

Answer: Working memory is the active *holding and manipulation* of information for ongoing cognitive tasks (reasoning, mental arithmetic, comprehension). It differs from simple short-term memory by emphasising *processing*, not just storage. Baddeley's model gives working memory multiple components (central executive + phonological loop + visuospatial sketchpad + episodic buffer), while STM in earlier models was just a passive buffer. Example: repeating a phone number = STM; computing a tip on a restaurant bill without paper = working memory.

Q3. [Essay] Describe the Atkinson–Shiffrin multi-store model. What process moves information between each stage?

Answer: The Atkinson–Shiffrin model proposes three sequential stores: **Sensory Memory** → **Short-Term Memory** → **Long-Term Memory**. Sensory memory briefly holds all sensory input (iconic, echoic); **attention** selects items for transfer to STM. STM has limited capacity (7 ± 2 chunks) and short duration (~30s) without rehearsal. **Rehearsal** transfers STM contents to LTM (large, durable). **Retrieval** pulls LTM contents back into STM for use. Information not rehearsed decays from STM; information not consolidated does not reach LTM. Limitations of the model: oversimplifies STM (Baddeley showed it has multiple components); does not capture the richness of working memory.

Q4. [Multiple Choice] The process of transferring information from short-term memory to long-term memory is called:

- (A) Retrieval (B) Encoding (C) Consolidation (D) Decoding

Answer: C. Consolidation stabilises and transfers labile STM traces into durable LTM storage. Involves synaptic consolidation (hours, protein synthesis) and systems consolidation (days to years, shift from hippocampus to cortex).

Q5. [Short Answer] Explain context-dependent learning with an example.

Answer: Context-dependent learning: information is better recalled when the

retrieval context matches the encoding context. Example: students who study and take an exam in the same room often perform better than students who study elsewhere; the room's sights, smells, and spatial layout serve as retrieval cues. Related finding (Godden & Baddeley): divers who learned word lists underwater recalled them better underwater than on land. Practical tip: to free your memory from specific contexts, *vary* your study environment.

- Q6. [Essay]** Describe Baddeley's working memory model. Name and explain each of its four components.

Answer: Baddeley's model has four components: (1) **Central executive** — the attentional controller; directs focus and coordinates the slave systems; resides in prefrontal cortex. (2) **Phonological loop** — stores and rehearses verbal/acoustic information (inner voice); used when you repeat a phone number silently. (3) **Visuospatial sketchpad** — stores and manipulates visual/spatial info (mentally rotating an object, visualising a room). (4) **Episodic buffer** (added 2000) — integrates information across modalities into coherent episodes; serves as an interface between working memory and long-term memory. Evidence for separate components: dual-task experiments show that verbal and visuospatial tasks interfere with themselves but not with each other.

- Q7. [Multiple Choice]** The ability to recognise and recall information *without conscious effort* (e.g., skills and habits) is called:

(A) Implicit (non-declarative) memory (B) Explicit (declarative) memory
(C) Episodic memory (D) Working memory

Answer: **A.** Implicit memory (procedural, priming) operates without conscious awareness. Explicit/declarative memory requires conscious recall.

- Q8. [Short Answer]** What is chunking and how does it overcome the limits of working memory?

Answer: **Chunking** is the grouping of individual items into larger, meaningful units. Since working memory capacity is $\sim 7 \pm 2$ *chunks* (not raw items), combining items into fewer chunks effectively increases capacity. Example: the string IBCIIITHYDMRMIQ is 16 characters (exceeds limit) but can be chunked as IBC | IIIT | HYD | MRI | MIQ = 5 chunks (within limit). Expert chess players chunk board positions into familiar patterns, holding far more pieces than novices within the same working memory capacity.

- Q9. [Essay]** Describe the case of Patient H.M. and explain what it revealed about the organisation of memory.

Answer: H.M. (Henry Molaison) had the bilateral medial temporal lobes (including hippocampus) surgically removed in 1953 to treat intractable epilepsy. Outcome: cured his seizures but produced profound **anterograde amnesia** — he could not form new declarative memories for the rest of his life. However: (1) his **old** memories (pre-1953) were intact (dissociating encoding from storage); (2) his **working memory** was normal (could hold info briefly); (3) his **proce-**

dural memory was preserved (he improved at mirror-tracing over days, though he never remembered having practised). **Implications:** (a) the hippocampus is essential for forming new declarative memories; (b) memory systems are *modular* and separable (declarative vs. procedural); (c) STM/working memory uses different circuits from LTM.

Q10. [Short Answer] What is the difference between shallow and deep processing? Which produces better memory retention?

Answer: **Shallow processing** focuses on surface features (font, sound, acoustic properties). **Deep processing** focuses on semantic content (meaning, connections to existing knowledge). **Deep processing produces substantially better retention.** Mechanism: semantic encoding creates richer associations in LTM, providing more retrieval paths. Study tip: ask *why* and *how* a fact is true, connect it to what you already know — this outperforms mere repetition.

Lecture 11: Memory Part 2 — Recall, Consolidation, and Distortion (April 17)

Supplementary PDF

The slide/handwritten PDF titled **Memory IBC part 2.pdf** is in the **April-17/** folder. Refer to it for diagrams of the forgetting curve and consolidation models.

11.1 Memory Recall is Reconstructive

Memory is Not a Recording

A crucial insight: memory recall is **not** like playing back a tape. Each retrieval is an active *reconstruction*, influenced by:

- Current cognitive load and emotional state
- Retrieval context and cues
- Prior expectations, biases, and schemas
- Previous acts of retrieval (each recall can modify the trace)

This reconstructive nature explains why memories can distort, shift, and even become false over time.

11.2 Familiarity vs. Recognition

Familiarity and Recognition

- **Familiarity** — a vague feeling of “knowing” something without being able to recall specific details (“I’ve met this person before but can’t remember where/when”)
- **Recognition** — specific identification of something as previously encountered, with access to contextual details

Neural Substrate of Familiarity

Familiarity and recognition rely on partly *separate* medial-temporal-lobe substrates:

- **Medial (perirhinal) cortex** — supports *familiarity* signals
- **Hippocampus** — supports *recollection* (episodic recall with contextual detail)

When you encounter a familiar stimulus, the medial cortex activates, generating the “I’ve seen this” feeling, even when the hippocampus cannot supply the when/where context.

11.3 Episodic vs. Semantic Memory (Deeper)

Episodic vs. Semantic

- **Episodic memory** — personally experienced events tied to a specific time and place. You can mentally “re-experience” them. Example: your last birthday party.
- **Semantic memory** — general factual knowledge, decontextualised. Example: the capital of France is Paris.

From Episodic to Semantic

Over time, episodic memories can become *semanticised*: you may remember that you once broke your arm as a fact, without being able to re-experience the episode. This transformation is part of systems consolidation — the hippocampus becomes less necessary as the memory is gradually stored in the neocortex.

11.4 Synaptic Plasticity — The Cellular Basis of Memory

Synaptic Plasticity

Synaptic plasticity is the ability of synaptic connections to change strength in response to activity. It is the *cellular basis of learning and memory*.

LTP, LTD, and Hebbian Learning

- **Long-Term Potentiation (LTP)** — persistent strengthening of synapses that fire together
- **Long-Term Depression (LTD)** — persistent weakening of synapses when firing is uncorrelated
- **Hebbian rule** (Donald Hebb, 1949): “*Neurons that fire together, wire together.*”

LTP was first identified in the hippocampus and is the most-studied candidate mechanism for memory formation. Protein synthesis is required to make LTP persistent (converts from *early* LTP to *late* LTP over hours).

11.5 Sleep and Memory Consolidation

Why Sleep Matters for Memory

Sleep is *essential* for memory consolidation. During sleep, the brain actively **replays** neural activity patterns from daytime experience, strengthening the associated synaptic connections and transferring memories from hippocampus to neocortex.

Key sleep stages:

- **Slow-wave (deep) sleep** — declarative memory consolidation
- **REM sleep** — procedural and emotional memory consolidation

Quality of sleep (proper cycling through stages) matters more than duration alone. Sleep deprivation impairs consolidation.

The Rat Maze Replay Study

A landmark finding: rats trained to navigate a maze show specific “place cell” firing sequences corresponding to their trajectory. During subsequent sleep, the *same* place cells fire in the *same sequence* (often compressed in time) — the brain is “replaying” the maze run.

This replay **strengthens** the memory. Rats with disrupted sleep replay show worse maze performance the next day. The finding generalises to humans (fMRI studies show similar replay of hippocampal activity during sleep). Upshot: pulling an all-nighter before an exam sabotages the very process that would have consolidated what you studied.

11.6 Context-Dependent Memory (Deeper)

Context Noise and Multiple Contexts

- When you study in a *single context*, memories are tightly bound to that context. Retrieval outside that context suffers (**context-dependent forgetting**).
- When multiple memories share the same context, they interfere — called **context noise**.
- **Multi-context learning**: studying across different contexts (different rooms, times of day) creates multiple retrieval paths to the same information, making it more robust.
- Actively *thinking about* differences between contexts further strengthens learning.

Practical implication: vary your study environment rather than always studying in the same spot.

11.7 Memory Distortion and False Memories

How False Memories Arise

Memories can become distorted or entirely false because:

1. The brain tries to *explain* partial recall, filling in gaps with plausible content
2. Random neural activity can resemble activity patterns of true memories
3. Suggestions from others (“didn’t you see X?”) can be incorporated into memory
4. Each retrieval re-encodes a (possibly modified) version of the memory

Forensic implications: eyewitness testimony is notoriously unreliable — leading questions can implant false memories. Elizabeth Loftus’s lost-in-the-mall studies showed false childhood memories can be induced in adults via suggestion.

11.8 Forgetting Curve — Ebbinghaus

Ebbinghaus’s Forgetting Curve

Hermann Ebbinghaus (1885) pioneered experimental memory research using **non-sense syllables** (“WUX,” “CAZ,” etc.) to eliminate prior-knowledge confounds. He tested his own retention over time.

Findings:

- Forgetting is *rapid at first*, then slows (exponential/power-law decay)
- ~60% of newly learned items are forgotten within 20 minutes
- By 24 hours, ~70% is forgotten
- After a week, what remains is relatively stable

Spaced repetition (reviewing at expanding intervals: 1 day, 3 days, 1 week, 2 weeks...) counteracts the forgetting curve. Modern flashcard apps (Anki, Quizlet) implement algorithmic spaced repetition.

11.9 Retrieval — Free Recall, Cued Recall, Recognition

Three Forms of Retrieval

- **Free recall** — retrieve without any cues (“list everything you studied”)
- **Cued recall** — retrieve given a prompt (“what was the word starting with ‘A’?”)
- **Recognition** — judge whether an item was previously encountered (“is this on the list?”)

Difficulty ranking: free recall > cued recall > recognition (easiest). A multiple-choice exam tests recognition; an essay exam tests free recall.

11.10 Modularity in Memory Storage

Memories are Stored Where They Are Used

Different memory types are stored in different brain regions appropriate to their content:

- **Motor skills** (procedural) — motor cortex, cerebellum, basal ganglia
- **Facts and concepts** (semantic) — temporal lobe, association cortices
- **Personal events** (episodic) — hippocampus + neocortex (over time)
- **Emotional associations** — amygdala
- **Visual features** — visual cortex (occipital)
- **Language content** — left temporal lobe (Wernicke’s area and surrounds)

A single “episode” is reconstructed by reactivating many regions together. Damage to one region produces *selective* memory deficits.

11.11 Emotion and Memory

Emotional Memories are Stronger

Emotionally charged events are remembered more vividly and persistently than neutral events. Mechanism: the **amygdala** (emotion processing) modulates **hippocampal** consolidation, adding emotional “tags” that boost encoding.

Flashbulb memories: vivid, detailed memories of where one was during major emotional events (e.g., “where were you on 9/11?”). Historically thought to be perfectly accurate; research now shows they are just as reconstructive as other memories — but more confidently believed.

PTSD and over-consolidation: in trauma, the amygdala over-activates encoding, creating intrusive, vivid memories that are difficult to suppress. Treatments (exposure therapy, EMDR, propranolol) aim to allow re-consolidation under safer conditions.

Practice & Review

Q1. [Multiple Choice] Which brain region is primarily associated with *familiarity* signals (the feeling of knowing without specific recall)?

- (A) Hippocampus (B) Medial (perirhinal) cortex (C) Amygdala (D) Cerebellum

Answer: B. The medial/perirhinal cortex supports familiarity, while the hippocampus supports *recollection* (episodic recall with detail). Both are in the medial temporal lobe but functionally dissociable.

Q2. [Short Answer] What is the distinction between true memories and false memories? How can false memories form?

Answer: True memories correspond to actual past experiences. **False memories** are recollections that did not actually happen (or are significantly distorted). Mechanisms by which false memories form: (1) the brain fills gaps during retrieval with plausible content, sometimes in error; (2) random neural activity can resemble the patterns of true memories; (3) suggestions or leading questions can be incorporated into memory (eyewitness distortion; Loftus’s “lost in the mall”); (4) each retrieval re-encodes a possibly modified trace. The reconstructive nature of memory makes distortion intrinsic.

Q3. [Essay] Describe how sleep supports memory consolidation, using the rat maze replay study as evidence.

Answer: During sleep, the brain *replays* neural activity patterns from daytime experience, strengthening the synaptic connections that encoded those experiences. The classic rat maze study: rats trained to run a maze show specific hippocampal “place cell” firing sequences. During subsequent sleep (particularly slow-wave sleep), the *same* place cells fire in the *same sequence* (often time-compressed). Disrupting this replay impairs next-day maze performance. The effect generalises: slow-wave sleep consolidates declarative memories; REM sleep consolidates pro-

cedural and emotional memories. Implication: quality sleep is not optional for learning; sleep deprivation sabotages the consolidation process that turns short-term into long-term memory.

- Q4. [Multiple Choice]** The process of actively rebuilding a memory during retrieval, potentially introducing distortions, is called:

(A) Memory consolidation (B) Memory reconstruction
(C) Encoding (D) Priming

Answer: B. Memory reconstruction captures the active, inferential nature of recall. Each retrieval is a reconstruction, not a playback; this explains distortion and the formation of false memories over time.

- Q5. [Short Answer]** What is the difference between episodic and semantic memory? How does a memory move from episodic to semantic over time?

Answer: Episodic memory is personally experienced events tied to a specific time and place, mentally re-experienceable. **Semantic** memory is decontextualised factual knowledge. *Transformation:* through *systems consolidation* over days-to-years, memories originally dependent on the hippocampus gradually become integrated into distributed neocortical stores. The time/place context often fades first, leaving the decontextualised fact. Example: you might remember that you once broke your arm (fact), without being able to re-experience the episode (lost context).

- Q6. [Essay]** Explain context-dependent memory and context noise. Why might studying in multiple locations be better than studying in one?

Answer: Context-dependent memory: memories are bound to the context (physical, emotional, physiological) in which they were encoded; retrieval is stronger when context matches. **Context noise:** when multiple memories share the same context, they interfere, making specific retrieval harder. **Why multiple contexts help:** studying in different rooms, at different times, with different background music creates *multiple retrieval paths* to the same information. This reduces dependence on any single context and provides more cues for recall. Actively thinking about what is different between contexts further enhances this effect. Practical tip: vary your study environment; mix locations, furniture, and study conditions.

- Q7. [Multiple Choice]** The type of retrieval where an individual must produce memories *without external cues* is called:

(A) Free recall (B) Recognition (C) Cued recall (D) Priming

Answer: A. Free recall is hardest. Recognition (“is this on the list?”) is easiest; cued recall sits in between. An essay exam tests free recall; a multiple-choice exam tests recognition.

- Q8. [Short Answer]** Why did Ebbinghaus use nonsense syllables in his memory research? What did he discover about forgetting?

Answer: Ebbinghaus used **nonsense syllables** (pronounceable but meaningless,

e.g., WUX, ZAB) to eliminate the confound of prior semantic knowledge: real words might be remembered through associations, but nonsense items require fresh learning. Each item is neutral and comparable. **Discovery:** the *forgetting curve* — forgetting is rapid at first (60% within 20 minutes, 70% by 24 hours) and then slows (exponential/power-law decay). What remains after a week is relatively stable. Modern implication: spaced repetition (reviewing at expanding intervals) counteracts the curve and is more efficient than massed practice.

- Q9. [Essay]** Explain modularity in memory storage. Why is damage to one brain area associated with *selective* memory deficits rather than total memory loss?

Answer: **Modularity** in memory: different types of memory are stored in different brain regions appropriate to their content — motor skills in motor cortex/cerebellum/basal ganglia, facts in temporal lobe and association cortices, personal events in hippocampus + neocortex, emotional associations in amygdala, visual features in visual cortex, language in left temporal lobe. A memory of “my last birthday party” is a distributed reconstruction: the hippocampus binds together activations in visual cortex (visual scene), auditory cortex (music), amygdala (joyful feeling), and frontal lobe (knowing who was there). **Why selective deficits:** damage to one module removes its content contribution but spares the others. Examples: hippocampal damage → inability to form new declarative memories but procedural memory intact (Patient H.M.); amygdala damage → loss of emotional memory tagging but factual recall preserved; Broca’s area damage → speech production lost, comprehension intact. This modular organisation is why memory loss is rarely all-or-nothing.

- Q10. [Short Answer]** Describe Hebb’s rule and its relationship to LTP (long-term potentiation).

Answer: **Hebb’s rule** (Donald Hebb, 1949): “*Neurons that fire together, wire together.*” Formally: if neuron A repeatedly contributes to firing neuron B, the synaptic connection from A to B strengthens. **LTP** (long-term potentiation) is the molecular implementation: correlated pre- and post-synaptic activity triggers NMDA-receptor-dependent processes that strengthen the synapse persistently (lasting hours to days; with protein synthesis, indefinitely). LTP is the most-studied candidate neural substrate for learning. Discovered originally in the hippocampus; now shown across the brain. The complementary process, **LTD** (long-term depression), weakens uncorrelated synapses. Together, LTP and LTD implement the biological Hebb rule.

Lecture 12: The Musical Brain (April 21)

Supplementary PDF

The PDF `The Musical Brain.pdf` (in the `April-21/` folder) contains the slide deck for this lecture. Refer to it for diagrams of the auditory pathway, Hargreaves reciprocal feedback model, and Russell circumplex model.

12.1 Music as a Stimulus

Music is a **single stream of audio** that the auditory system receives and that the brain *separates* into constituent acoustic features, then *integrates* into higher-level percepts.

Auditory Scene

An **auditory scene** is the full acoustic environment heard at any moment. The brain parses it into separate *auditory streams* (e.g., separating a cello from background speech). This is the auditory analogue of visual scene parsing.

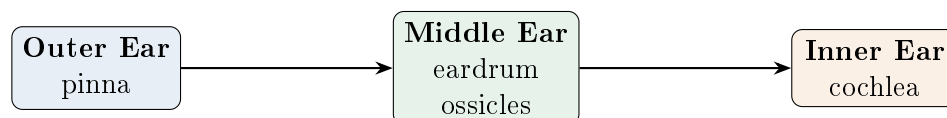
Music engages all four dimensions of the cognitive system simultaneously:

- **Sensory** — basic acoustic feature detection
- **Attention** — selective focus on one stream (melody, rhythm, instrument)
- **Memory** — recognition, emotional associations, prior exposure
- **Emotional** — arousal, valence, mood regulation

Music also provides real-world **affordances**: emotional regulation, sensory modulation, reflection and meaning-making, pleasure, and social connection and empathy. “*Music is a mirror of the self. Music is an expression of who we are emotionally, socially, and cognitively*” — Greenberg et al. (2015).

12.2 Anatomy of Hearing

Sound waves travel through three stages before reaching the brain:



Auditory Pathway to Cortex

From the cochlea, signals travel up the **subcortical auditory pathway**:

Cochlear Nucleus → Superior Olive → Inferior Colliculus → Medial Geniculate (Thalamus) → Primary Auditory Cortex

The **primary auditory cortex** is located in the **temporal lobe** (Heschl's gyrus). It performs initial frequency analysis; surrounding belt regions handle complex sound patterns including speech and music.

12.3 Sound Decomposition and Integration

How the Brain Processes Music

The audio signal is first **separated** into low-level acoustic features:

- **Pitch** — the perceived frequency of a note
- **Timbre** — the “colour” of a sound; what makes a violin different from a piano at the same pitch
- **Timing / Rhythm** — temporal structure of notes
- **Harmonics** — overtones above the fundamental frequency
- **Intensity / Loudness** — amplitude

These features are then **integrated** into higher-level musical constructs: *Melody, Harmony, Rhythm, Key, Genre, Emotional meaning, Musical identity/memories*.

Time-Scale Hierarchy of Music Recognition

Music is processed hierarchically across time scales (Rauschecker):

1. **0.1 s** — Pitch, Timbre, Loudness (basic acoustic features)
2. **1 s** — Pulse (beat), Streams (auditory scene analysis)
3. **10 s** — Melody, Instruments, Harmony
4. **100 s** — Genre, Emotion (global musical meaning)

Even a few seconds of music can trigger genre and artist recognition because the brain rapidly ascends this hierarchy.

12.4 Neural Pathways for Music Processing

Music processing relies on **dorsal** and **ventral** auditory streams, analogous to the “where” and “what” pathways in vision:

- **AC** (auditory cortex) — primary and belt areas, initial spectral analysis
- **STS** (superior temporal sulcus) — complex auditory patterns, speech, and music
- **IFC** (inferior frontal cortex / Broca’s area) — syntax, expectation, musical structure
- **PMC** (premotor cortex) — rhythm processing, entrainment, sensorimotor coupling
- **IPL** (inferior parietal lobule) — spatial and higher-order integration
- **CS** (central sulcus) — motor-auditory coupling

The dorsal pathway (AC → PMC) handles spatial and rhythmic processing; the ventral pathway (AC → STS → IFC) handles object identity (what instrument, what melody, what emotion).

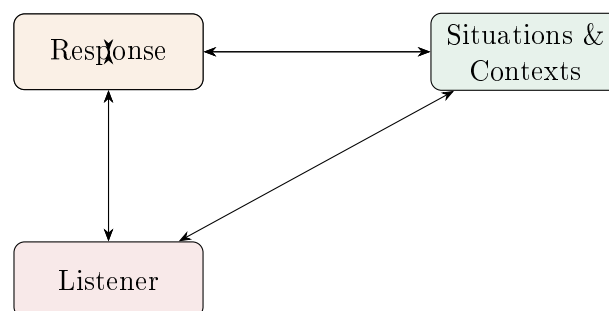
12.5 Hargreaves et al. (2005) Reciprocal Feedback Model

Reciprocal Feedback Model of Musical Response

Hargreaves et al. (2005) proposed that music response is determined by *three interacting factors*:

1. **Music** — reference system (genre, idiom), collative variables (complexity, familiarity, orderliness), prototypicality, performance context (live, recorded)
2. **Listener** — individual difference variables (gender, age, nationality), musical knowledge/training/literacy/experience, taste patterns, musical identity (self-theories)
3. **Situations and Contexts** — social/cultural context, everyday situations (work, leisure, entertainment), presence/absence of others

These three interact with a central **Response** node via *bidirectional* arrows.



Types of Musical Response

Responses to music are classified as:

- **Physiological** — arousal level, active/passive control of listening, heart rate
- **Cognitive** — attention, memory, perceptual coding, expectation, discrimination, evaluation
- **Affective** — emotional responses, like/dislike, mood

Listener Attribute

A **listener attribute** is any characteristic *unique to an individual* that makes them perceive or respond to music differently from another person. Includes:

- Personality traits (extraversion, neuroticism, openness)
- Musical knowledge, training, literacy, instrument experience
- Musical identity and self-theories (“I am a jazz person”)
- Cultural and social background

Key exam point: the same piece of music can elicit entirely different emotional responses in two people due to differing listener attributes.

12.6 Quantifying Emotional Response to Music

12.6.1 Russell’s Circumplex Model

Russell’s Circumplex Model of Emotion

Emotions are mapped onto a **2D plane** with two orthogonal axes:

- **Valence** (x-axis): from negative (0) to positive (1) — how pleasant/unpleasant
- **Arousal** (y-axis): from calm (0) to excited (1) — how activated/deactivated

Four quadrants:

1. **Q1** (High Arousal, High Valence) — **Happiness** / **Energy**
2. **Q2** (High Arousal, Low Valence) — **Anger** / **Tension** / **Fear**
3. **Q3** (Low Arousal, Low Valence) — **Sadness** / **Tiredness**
4. **Q4** (Low Arousal, High Valence) — **Tenderness** / **Calmness**

Thayer’s model adds dashed axes at 45° to capture tension–energy and tiredness–calmness dimensions. Basic emotion labels (fear, anger, disgust, sadness, tenderness, surprise) can all be placed on this plane.

Why the Circumplex Model is Useful

It provides a *universal, quantitative* framework for emotion: rather than debating which emotion label to apply, you simply map onto (Valence, Arousal) coordinates. Spotify and streaming services use this to create mood-based playlists by mapping songs’ acoustic features into this space.

12.6.2 Three Types of Emotional Response

Music-induced vs. Perceived vs. Aesthetic Response

For any musical stimulus, three *distinct* emotional responses can occur simultaneously:

1. **Music-induced emotions** — the emotion *actually felt* by the listener (subjective, felt response)
2. **Perceived emotional expression** — the emotion the listener *perceives the music to express* (not necessarily felt; a minor-key song can be perceived as “sad” without making the listener feel sad)
3. **Aesthetic responses** — appreciation of beauty/craft independent of personal emotion

These three can differ: a sad song may be perceived as expressing sadness (perceived), actually make you feel nostalgic (induced), and simultaneously be appreciated as beautiful (aesthetic).

The full picture: **Musical Features + Listener Attributes + Situational Factors** → the overlap of music-induced emotions, perceived emotional expression, and aesthetic responses. This is the *reciprocal feedback model of affective response*.

12.7 Research Applications

12.7.1 Does Sadness Come from Lyrics or Music?

Source separation approach: take a song (with known VA coordinates), algorithmically separate the audio into individual tracks (guitar, drums, piano, vocals/lyrics), perform *text-based sentiment analysis* on lyrics, and map the lyrics’ sentiment onto the VA space. If lyrics’ VA coordinates match the song’s overall VA coordinates, lyrics drive the emotion; if not, the instrumentation/harmony is responsible.

12.7.2 What Makes Music Recognisable? (Neural Network Probe)

A neural network is trained to **classify songs by genre**. Its layers are then *probed* to ask what each layer “knows”:

- Early layers — basic acoustic features (timbre, pitch)
- Middle layers (layers ~5–6) — instruments are strongly represented (genre-informative)
- Late layers — genre-level abstract categories

Intervention: replace violin with guitar in input → observe which layers change. This mirrors how neuroscientists use fMRI to localise music processing in the brain, but is cheaper and more controllable.

12.7.3 Healthy vs. Unhealthy Music Engagement

Healthy Music Engagement

Healthy music engagement leads to positive well-being outcomes:

- Music-based interventions for recovery from neurological/clinical disorders
- Creating positive emotional states and increased energy
- Promoting well-being and regulating emotions

(Reybrouck 2020; Gustavson et al. 2021; Alluri et al. 2022; Tan et al. 2024)

Unhealthy / Maladaptive Music Engagement

Maladaptive music engagement is associated with negative outcomes:

- Associated with internalising disorders (anxiety, depression)
- Negative emotionality and increased odds of depressive symptoms
- Susceptibility to negative emotional contagion
- Difficulty managing emotions when in distress

Critical distinction: it is not the music itself that is harmful, but *how* the listener is using it. The same song can be used adaptively (for catharsis) or maladaptively (for rumination). “*Health is socially produced and environmentally-mediated*” — DeNora (2007).

12.7.4 Digital Footprints and Data-Driven Research

Modern music research uses **digital footprints** — the traces left on streaming platforms (Spotify, Last.fm, Apple Music, Deezer) and social media (Reddit, Twitter):

- **Semi-structured data** from streaming APIs: metadata, lyrics, **acoustic features** (tempo, key, loudness, danceability), user-generated **tags**
- Acoustic features are embedded using adjective/adverb embeddings, then mapped onto the VA (valence-arousal) space
- **Word Emotion Induction Model** (Warriner et al. 2013): assigns Valence, Arousal, and Dominance scores to words on 9-point scales (e.g., “sunshine”: V=8.1, A=5.3)
- **Social media** (Reddit): posts scraped from depression-related communities to distinguish *healthy* (relaxation, mood regulation, coping, music therapy) from *unhealthy* (depressive music, mood worsening, listening in the dark, addicted to music) music discourse (Singh et al. 2023, *PLoS ONE*)

12.7.5 Emotion Transition Probabilities

Using Spotify/Last.fm listening histories, researchers model **emotion transition probabilities**: how likely a person transitions from Q1 $\rightarrow$ Q3 (happiness to sadness) across listening sessions. Key finding:

- **Highly sensitive persons** (low EOE-External) tend to transition into Q3 (low arousal, low valence) and stay there — consistent with rumination
- Less sensitive persons cycle more freely across quadrants

Inertia in VA space (staying in same quadrant) is a potential marker of maladaptive engagement.

12.7.6 Experience Sampling Method (ESM)

To capture real-world, *in-the-moment* responses, researchers use **experience sampling**:

- Participants report at multiple time points (Time 1, 2, 3 ...) while listening
- Physiological measures added: Heart Rate Variability, Goal Relevance, Hedonic Shift
- Enables **longitudinal** study of how music use patterns evolve over time

12.7.7 Network Analyses

Network analysis models psychological variables (listener attributes, music use patterns, well-being) as **interacting nodes** with edges encoding conditional dependencies. Key variables:

- **K10** (Kessler Psychological Distress Scale) — anxiety and depression symptoms
- **IRI** (Interpersonal Reactivity Index) — empathy: PT (Perspective Taking), FS (Fantasy Seeking), EC (Empathic Concern), PD (Personal Distress)
- **SWLS** (Satisfaction with Life Scale)
- **SSQ** (Social Support Questionnaire)
- EOE-External / EOE-Internal — measures of music engagement type

Directed acyclic graphs (DAGs) test whether the data fit a hypothesised causal model (e.g., EOE-External $\rightarrow$ unhealthy engagement $\rightarrow$ K10 distress).

12.8 Summary: The Musical Brain Framework

Integrated Framework

1. **Signal level:** audio → pitch/timbre/timing/harmonics/intensity (separated) → melody/rhythm/harmony/genre (integrated)
2. **Anatomy:** outer ear → middle ear → cochlea → cochlear nucleus → superior olive → inferior colliculus → medial geniculate → primary auditory cortex
3. **Neural pathways:** dorsal (rhythm/spatial) and ventral (identity/meaning) auditory streams
4. **Response:** physiological + cognitive + affective; three types — music-induced, perceived emotional expression, aesthetic
5. **Model:** Hargreaves (2005) reciprocal feedback among Music ↔ Listener ↔ Contexts ↔ Response
6. **Measurement:** Russell circumplex (Valence × Arousal); four quadrants Q1–Q4
7. **Well-being:** healthy engagement (positive outcomes) vs. maladaptive engagement (rumination, internalising disorders)

Practice & Review

Q1. [Multiple Choice] In Hargreaves et al.’s (2005) reciprocal feedback model, which of the following is NOT one of the three main components?

(A) Music (B) Listener (C) Composer (D) Situations and Contexts

Answer: C. The three components are Music, Listener, and Situations/Contexts, all interacting with a central Response node. The composer’s intent is encoded in the music itself but is not a separate component of the response model.

Q2. [Short Answer] Distinguish between music-induced emotion, perceived emotional expression, and aesthetic response. Give an example.

Answer: Music-induced emotion: the emotion *actually felt* by the listener (e.g., feeling nostalgic). **Perceived emotional expression:** the emotion the listener perceives the music to be expressing (e.g., recognising a minor-key ballad as “sad” without feeling sad yourself). **Aesthetic response:** appreciation of the craft or beauty of the music independent of any specific emotion (e.g., appreciating Beethoven’s technical mastery while feeling neutral). Example: a friend’s wedding song may be perceived as joyful (perceived), induce sadness due to a breakup memory (induced), and simultaneously be appreciated as beautifully composed (aesthetic).

Q3. [Multiple Choice] In Russell’s circumplex model, which quadrant corresponds to high arousal and low valence?

(A) Q1 — Happiness (B) Q2 — Anger (C) Q3 — Sadness (D) Q4 — Tenderness

Answer: **B.** Q2 = High Arousal + Low (negative) Valence = Anger/Tension/Fear. Q1 = High Arousal + High Valence = Happiness. Q3 = Low Arousal + Low Valence = Sadness. Q4 = Low Arousal + High Valence = Tenderness/Calmness.

Q4. [Short Answer] What is a listener attribute? Give three examples.

Answer: A **listener attribute** is any characteristic unique to an individual that shapes how they perceive and respond to music, differently from others. Examples: (1) musical training (a classically trained pianist perceives harmony differently from a non-musician); (2) personality traits (high openness-to-experience correlates with liking complex music); (3) cultural background (familiarity with Carnatic vs. Western tonal systems affects what sounds “in tune”).

Q5. [Short Answer] Name the subcortical stations of the ascending auditory pathway from cochlea to cortex.

Answer: Cochlea → **Cochlear Nucleus** → **Superior Olive** → **Inferior Colliculus** → **Medial Geniculate Nucleus** (thalamus) → **Primary Auditory Cortex** (temporal lobe, Heschl’s gyrus).

Q6. [Essay] What is the difference between healthy and unhealthy music engagement? Why is it the *use* rather than the *music* that matters?

Answer: **Healthy music engagement** leads to positive outcomes: emotional regulation, reduced anxiety/depression symptoms, increased well-being, social connection. **Unhealthy/maladaptive engagement** is associated with internalising disorders, rumination (dwelling in Q3 — low arousal, low valence), negative emotional contagion, and difficulty managing distress. Why use not music: the same song (e.g., a slow minor-key ballad) can be used adaptively — to process and release sadness (catharsis) — or maladaptively — to wallow and ruminate. Individuals at risk of depression tend to use music to amplify negative emotion rather than regulate it. The music itself is neutral; listener intent and context determine outcome. This is consistent with DeNora’s (2007) principle: “health is socially produced and environmentally mediated.”

Q7. [Short Answer] What is the time-scale hierarchy of music recognition? Why can humans recognise a genre in under a second?

Answer: Music is processed in layers of increasing time scale: at 0.1 s — pitch, timbre, loudness; at 1 s — pulse (beat) and auditory streams; at 10 s — melody, instruments, harmony; at 100 s — genre and emotion. Recognition in under a second is possible because the brain immediately extracts timbre (instrument colour) and rhythmic pulse from the first few milliseconds — these alone are genre-diagnostic (e.g., a guitar riff vs. a string quartet). The hierarchy ascends rapidly due to parallel processing across auditory areas.

Lecture 13: Attention — Selection, Search, and Feature Integration (April 24)

Supplementary PDF

The PDF `Lecture - Attention Perception and Performance_v4.pdf` (in the `April-24/` folder) contains the slide deck for this lecture. Refer to it for figures of Treisman's feature integration model and attention paradigms.

13.1 What is Attention?

Attention

Attention is the cognitive process of *selectively focusing* on relevant information while *ignoring* irrelevant information. It is a **limited cognitive resource** allocated across tasks, stimuli, and mental operations.

Why Attention is Necessary

The sensory environment contains *far more* information than the brain can process consciously:

- Retinas deliver millions of bits per second; consciousness handles only tens of bits per second
- Multiple conversations, sounds, and visual events co-occur constantly
- Without selection, the system would be overwhelmed

Attention solves this by gating which inputs get deep processing. It is *central* to perception, memory, learning, and decision-making — attention determines what you will even have the chance to remember.

13.2 Types of Visual Search

Self-Terminating vs. Exhaustive Search

- **Self-terminating search:** searching stops as soon as the target is found (e.g., scanning a bookshelf for one specific book and stopping when found)
- **Exhaustive search:** all items are examined even after the target is found (e.g., checking every shelf to confirm no duplicates exist)

Reaction-Time Signatures

These two search types leave different reaction-time signatures:

- **Self-terminating:** search time is *shorter on target-present trials* than on target-absent trials (you stop early when you find it)
- **Exhaustive:** search time is *the same* for target-present and target-absent trials (you always finish)

Experimental paradigms use these signatures to infer which search mode the brain is using.

13.3 Selective Attention

Selective Attention

Selective attention is the ability to focus on one stimulus or task while ignoring competing stimuli. The classic example: the **cocktail party effect** — attending to one conversation in a noisy room and suppressing others (though your name spoken across the room can still capture attention).

- Essential for learning — unattended material is not encoded
- Essential for safe behaviour — driving requires attending to relevant cues and ignoring billboards
- Fails in *attentional disorders* such as ADHD

13.4 Attention as a Limited Resource

Resource Allocation

Attention is a *limited resource*. Performance depends on how well it is allocated:

- Doing two demanding tasks simultaneously usually degrades both (driving + texting)
- **Task switching** incurs a cost — every switch consumes time and energy
- With practice, previously demanding tasks become *automatic* and free up attention (expert driver, from Lecture 8's skill-learning stages)

Attention is the currency of the cognitive economy: you must spend it wisely.

13.5 Early vs. Late Selection — Filter Models

A central debate in attention research: *where* in processing does the filter operate?

13.5.1 Broadbent's Filter Model (Early Selection, 1958)

Broadbent — Early Filter

Donald Broadbent proposed attention acts as a *bottleneck* early in processing:

- All sensory inputs briefly enter a sensory buffer
- An *attention filter* selects one channel (based on physical features like ear or pitch) for further processing
- Unattended inputs are *blocked* entirely before semantic analysis

Problem with pure early selection: you still notice your own name in the unattended ear — but if blocked, semantic content could not be extracted.

13.5.2 Deutsch–Norman Late Selection

Late Selection

Deutsch and Norman proposed the opposite: *all* inputs are processed semantically, and selection occurs *late* (based on meaning) before entering awareness/memory.

- Explains why unattended-but-important content (your name) is detected
- Criticised as wasteful — too much processing of material that will be discarded

13.5.3 Treisman's Attenuation Model (Compromise)

Treisman — Attenuation, not Blocking

Anne Treisman proposed an intermediate solution: the filter *attenuates* (turns down) unattended channels rather than blocking them entirely.

- Unattended channels are processed at reduced intensity
- Salient or meaningful content (your name, a loud bang) can still “break through”
- Reconciles early and late selection evidence

13.6 Top-Down vs. Bottom-Up Attention

Top-Down vs. Bottom-Up Attention

- **Top-down (endogenous, goal-directed) attention** — voluntary, driven by current goals, prior knowledge, and expectations. Example: searching for your friend in a crowd — you bias processing toward their face features.
- **Bottom-up (exogenous, stimulus-driven) attention** — involuntary, captured by salient stimuli. Example: a sudden loud noise jerks your attention; a flashing light in peripheral vision draws your gaze.

Interaction of the Two

Both operate simultaneously:

- Top-down enhances goal-relevant processing (prefrontal cortex)
- Bottom-up interrupts when something unexpected happens (superior colliculus, parietal cortex)
- The balance is continuously negotiated — too much bottom-up makes you distractible; too much top-down makes you miss important novel events (“inattention blindness”)

13.7 Feature Integration Theory (Treisman)

Feature Integration Theory (FIT)

Anne Treisman’s influential theory (1980) proposes that visual perception has two stages:

1. **Preattentive stage** — features (colour, shape, orientation, motion) are processed *in parallel* across the visual field; fast and automatic
2. **Attentive stage** — *attention* binds these separate features into unified *objects*; serial and effortful

The Binding Problem

Binding problem: when you see a red apple, your visual system separately detects *red* (colour pathway) and *round* (shape pathway). How are these bound together into a single object percept rather than being misassigned (perceiving a green apple + red square)?

Treisman's answer: attention is the glue. Focused attention at a location binds features at that location into a unified object. Without attention, *illusory conjunctions* occur — the visual system mismatches features across objects.

Evidence: in brief displays with divided attention, participants report combinations that were not actually present (e.g., a red T and a green X displayed → reported as a green T and a red X).

13.8 Brain Areas Involved in Attention

Brain Region	Attention Role
Parietal Lobe	Spatial attention; disengaging and shifting attention; binds features into objects
Prefrontal Cortex	Executive/top-down attention; goal maintenance; conflict resolution
Visual Cortex	Modulated by attention — attended locations show enhanced responses
Thalamus (pulvinar)	Attention gating and selection across sensory modalities
Anterior Cingulate Cortex	Conflict monitoring; error detection
Superior Colliculus	Reflexive attention shifts; eye movements

Hemispatial Neglect

Patients with damage to the right parietal lobe (typically from stroke) often develop **hemispatial neglect**: they ignore the left half of space as if it does not exist.

- They eat only from the right half of their plate
- They shave/apply make-up only to the right side of their face
- When drawing a clock, they place all numbers on the right side
- They deny the deficit even when pointed out

Neglect demonstrates that the **right parietal lobe** is critical for attending to the *left* side of space. It is not a sensory deficit (vision is intact) — it is an *attentional* deficit. This case illustrates how attention is not a single monolithic process but is distributed across specific brain networks.

13.9 Attention vs. Intention

Attention and Intention

- **Intention** — the goal or purpose driving behaviour
- **Attention** — the selective focusing of cognitive resources

These often co-occur but are dissociable:

- *Attention with intention*: you search for your keys (goal) and attend to locations they might be
- *Attention without intention*: a sudden bang captures your attention involuntarily (bottom-up)
- *Intention without attention*: you want to write an email but keep getting distracted

13.10 Attentional Control

Attentional Control

Attentional control is the meta-ability to *regulate* one's own attention — directing it to relevant tasks and resisting distraction. It is a core component of executive function, mediated by the prefrontal cortex.

- Develops gradually throughout childhood; matures with the prefrontal cortex (~age 21)
- Impaired in ADHD
- Can be trained — meditation practitioners show enhanced attentional control
- Critical for academic performance (focus during study)

13.11 Cognitive Economy

Cognitive Economy

Cognitive economy refers to the efficient allocation of limited cognitive resources — attention, working memory, effort — to achieve goals. Our brains are miserly: they offload cognition wherever possible:

- Automatising frequent skills (driving, reading) to free up attention
- Using *heuristics* instead of full analysis for rapid decisions
- Chunking information to expand effective working memory
- Relying on schemas and scripts instead of re-analysing familiar situations

Understanding cognitive economy helps explain biases and errors: when the resource is expensive, shortcuts are used, sometimes at the cost of accuracy.

13.12 Attention in Real-World Contexts

- **Driving:** attention is allocated to road, mirrors, speedometer, other vehicles, pedestrians. Texting while driving fragments attention catastrophically
- **Learning:** students who attend deeply and avoid multitasking retain more. Cell-phone use during lectures has been shown to reduce exam performance
- **Sports:** expert athletes attend to task-relevant cues (ball, opponent) and filter out crowd noise
- **Meditation:** trains sustained and meta-attention; shown to modify attention networks

Practice & Review

Q1. [Multiple Choice] The primary function of attention in cognitive processing is to:

- (A) Process all incoming sensory information (B) Filter out irrelevant information and focus on relevant information
(C) Store information in long-term memory (D) Generate motor output

Answer: B. Attention acts as a selective gate, allowing only relevant information to receive deep processing and be encoded into memory. Without this selection the cognitive system would be overwhelmed.

Q2. [Short Answer] Distinguish between self-terminating and exhaustive search. Give an example of each and explain how their reaction-time signatures differ.

Answer: Self-terminating search: search stops as soon as the target is found.

Example: scanning a shelf for one specific book and stopping when found. **Exhaustive search:** all items are examined even after the target is found. Example: checking for all duplicates in a list. **RT signatures:** self-terminating searches are *faster on target-present trials* than target-absent trials (you quit early when you find it); exhaustive searches have *equal RT* for both (you always finish the scan). These signatures let researchers infer which mode the brain is using.

- Q3. [Essay]** Compare Broadbent’s early-selection model, Deutsch–Norman’s late-selection model, and Treisman’s attenuation model of attention.

Answer: **Broadbent’s early-selection** (1958): attention acts as a bottleneck *early* in processing; an attention filter selects one channel based on physical features (ear, pitch); unattended inputs are *blocked* before semantic analysis. Problem: we still hear our name in the unattended ear, which requires some semantic processing. **Deutsch–Norman late-selection:** *all* inputs are processed semantically; selection occurs *late* based on meaning before entering awareness. Explains the name-detection but is wasteful. **Treisman’s attenuation:** the filter *attenuates* (turns down) unattended channels rather than blocking them. Salient content (your name, loud bang) can still break through at reduced threshold. Reconciles early and late evidence. Modern view: attention operates at multiple levels, with flexible gating depending on task demands.

- Q4. [Multiple Choice]** Which brain area is most critical for *spatial* attention, as evidenced by hemispatial neglect?

(A) Cerebellum (B) Parietal lobe (C) Temporal lobe (D) Brainstem

Answer: **B.** Damage to the right parietal lobe causes left hemispatial neglect — patients ignore the left half of space despite intact vision. This shows the right parietal lobe is critical for attending to the left side of space.

- Q5. [Short Answer]** Describe the binding problem in visual perception. How does Feature Integration Theory solve it?

Answer: The **binding problem:** the visual system processes features (colour, shape, orientation, motion) in *separate parallel pathways*. When you see a red apple, “red” and “round” are detected separately. How are they bound together into a unified object so you don’t perceive the red applied to the wrong shape? **Treisman’s FIT solution:** attention is the glue. At the **preattentive stage**, features are registered in parallel maps without conjunction. At the **attentive stage**, focused spatial attention binds features at the attended location into unified objects. Evidence: under divided attention, *illusory conjunctions* occur — participants mis-bind features across objects (seeing a red T when a red X and green T were present).

- Q6. [Essay]** Explain top-down versus bottom-up attention. Give examples of each and describe a situation where both operate simultaneously.

Answer: **Top-down attention** (endogenous, goal-directed, voluntary): driven by current goals, prior knowledge, and expectations. Mediated by prefrontal cortex.

Example: searching for your friend in a crowd — you voluntarily bias processing toward their face features. **Bottom-up attention** (exogenous, stimulus-driven, involuntary): captured by salient stimuli regardless of goals. Mediated by parietal cortex and superior colliculus. Example: a sudden loud bang jerks your attention; a flashing light captures your gaze. **Simultaneous operation**: driving a car — top-down attention keeps you oriented to the lane ahead and your destination, while bottom-up attention catches an unexpected pedestrian stepping from the sidewalk. If top-down dominates too much, you miss the pedestrian (inattentional blindness); if bottom-up dominates, you are distracted by every billboard. The balance is continuously negotiated.

Q7. [Multiple Choice] The prefrontal cortex is most involved in which type of attention?

- (A) Reflexive (bottom-up) attention (B) Spatial orienting
(C) Executive (top-down) attention (D) Peripheral visual attention

Answer: C. Prefrontal cortex maintains goals, resolves conflicts, and directs top-down executive attention. Reflexive/bottom-up attention relies on the parietal cortex and superior colliculus; spatial orienting involves parietal areas.

Q8. [Short Answer] What is the cocktail party effect? Which type of attention does it illustrate?

Answer: The **cocktail party effect**: the ability to focus on one conversation at a noisy party while filtering out competing conversations. It illustrates **selective attention** — choosing one channel to process deeply while suppressing others. A key related observation: you can still detect your own name in the unattended channel, showing that unattended input is not completely blocked but only attenuated (Treisman's view).

Q9. [Essay] Describe hemispatial neglect. Why is it an *attentional* rather than a *visual* deficit?

Answer: **Hemispatial neglect**: following damage (usually stroke) to the right parietal lobe, patients fail to attend to the left half of space. They eat only from the right side of their plate, shave only the right side of their face, place all clock numbers on the right side when drawing, and often deny the deficit. **Why it is attentional, not visual**: their visual system is intact (the left-visual-field retinotopy to occipital cortex is unimpaired). If shown a stimulus on the left with their attention drawn to it, they can often detect it. The deficit is in *orienting* attention leftward — the brain region that normally directs spatial attention to the left (right parietal cortex) is damaged. This dissociation is clinically important: rehabilitation programmes train left-side attention, not the visual system itself. Neglect also demonstrates the modular nature of attention — different attention processes rely on different brain circuits.

Q10. [Short Answer] Explain cognitive economy. Give two examples of how the brain minimises cognitive effort.

Answer: **Cognitive economy** is the efficient allocation of limited cognitive resources (attention, working memory, effort) to achieve goals. The brain offloads expensive cognition wherever possible. Examples: (1) **Automaticity** — with practice, formerly effortful skills (driving, reading) become automatic, freeing attention for other tasks. (2) **Heuristics** — fast rules-of-thumb replace full analysis for rapid decisions (e.g., availability heuristic: estimate frequency by how easily examples come to mind). (3) **Chunking** — grouping items into larger meaningful units to expand effective working memory capacity. (4) **Schemas/scripts** — using stored frameworks to navigate familiar situations (restaurant schema) without re-analysing from scratch. Cognitive economy explains both our efficiency and our biases — shortcuts are fast but sometimes wrong.

Comprehensive Summary

Supplementary: Handwritten Notes and Slide Decks

Several supplementary PDFs accompany this guide:

- `Notes/IBC-26_Mar_2026.pdf` — handwritten notes from the March 26 session
- `April-10/IBC_Intelligence_10Apr2026.pdf` — Intelligence lecture slides
- `April-14/Memory IBC 2026 part 1.pdf` — Memory Part 1 slides/notes
- `April-17/Memory IBC part 2.pdf` — Memory Part 2 slides/notes
- `April-24/Lecture - Attention Perception and Performance_v4.pdf` — Attention lecture slides

Refer to these for visual diagrams (Gardner’s multiple intelligences, Atkinson–Shiffrin and Baddeley memory models, Ebbinghaus forgetting curve, Treisman’s feature integration theory) and additional detail that complements this typed guide.

Timeline of Key Ideas

Era	Approach	Central Thesis
1800s	Introspection	Self-observation reveals the mind
1900s–1950s	Behaviourism	Only S–R associations; no “mind”
1950s–60s	Cognitive Revolution	Mind mediates behaviour; internal representations
Emerging	Consciousness studies	Qualia; subjective experience; self-awareness

Brain Organisation at a Glance

Region	Key Functions
Frontal Lobe	Executive function, motor control, speech (Broca's), planning, personality
Parietal Lobe	Somatosensory, proprioception, spatial orientation
Occipital Lobe	Visual processing (object recognition, motion)
Temporal Lobe	Auditory processing, language comprehension (Wernicke's), object naming
Insula	Sense of self, saliency processing
Amygdala	Fear, emotional evaluation
Hippocampus	Long-term memory
Basal Ganglia	Motor gateway
Thalamus	Sensory relay station
Hypothalamus	Homeostasis (hunger, temperature, breathing)
Cerebellum	Motor coordination, balance
Brainstem	Vital functions (respiration, heart rate, sleep–wake)

Key Methods Summary

Method	Measures	Invasive?	Temporal Res.
EEG	Electrical activity (scalp)	No	ms (excellent)
MEG	Magnetic fields (scalp)	No	ms (excellent)
fMRI	Blood oxygenation (BOLD)	No	seconds (poor)
PET	Glucose metabolism	Yes (radiotracer)	minutes (poor)
ECoG	Electrical activity (cortex)	Yes (surgery)	ms (excellent)
DBS	Stimulates subcortical areas	Yes (surgery)	—

Key Numerical Values to Remember

Quantity	Value
Working memory capacity	7 ± 2 chunks
Number of neurons in the brain	$\sim 10^{10}$ (10 billion)
Average connectivity per neuron	$\sim 10^4$ (10,000 synapses)
Simple visual reaction time	~ 100 ms
Object recognition threshold	~ 100 ms
Batsman response time needed	$\sim 15\text{--}50$ ms
Action potential duration	~ 1 ms
Action potential amplitude	~ 30 mV (from -70 to $+30$ mV)
Resting membrane potential	-70 mV
Firing threshold	-55 mV
Synaptic cleft width	~ 1 μm
CNS maturation age	~ 21 years
Frontal lobe maturation	$\sim 18\text{--}21$ years
ERP signal amplitude	~ 2 μV
Facial muscles	~ 200
Sensory memory duration	$\sim$ seconds
Short-term memory duration	$\sim$ minutes to hours
Skill learning (bicycle)	~ 1 month practice
Expert driving acquisition	~ 2 months practice

Important Case Studies

1. **Phineas Gage** — Iron rod through frontal lobe $\rightarrow$ personality change, loss of planning/social behaviour
2. **Broca's Aphasia** — Frontal lobe damage $\rightarrow$ cannot produce speech, can comprehend
3. **Wernicke's Aphasia** — Temporal lobe damage $\rightarrow$ fluent but nonsensical speech, cannot comprehend
4. **Capgras Syndrome** — Intact recognition but lost emotional evaluation $\rightarrow$ "imposter" delusion
5. **Alzheimer's Disease** — Temporal lobe neurodegeneration $\rightarrow$ memory loss, disorientation; procedural memory remains intact
6. **Parkinson's Disease** — Basal ganglia (subthalamic nucleus) dysfunction $\rightarrow$ resting tremor, rigidity; treated with DBS (dramatic video demonstration)

7. **Cerebellar Ataxia** — Cerebellum damage (e.g., chronic alcoholism) → loss of motor coordination
8. **Hemiplegia** — Motor output cut (e.g., stroke) → motor intention intact but cannot execute; target for BCI rehabilitation
9. **Visual Illusions** — Kanizsa triangle, cafe wall illusion demonstrate bottom-up perceptual construction
10. **Patient H.M. (Henry Molaison)** — Bilateral medial temporal lobe removal → profound anterograde amnesia; procedural memory and old memories intact. Proved the hippocampus is essential for forming new declarative memories and that memory systems are modular
11. **Rat Maze Replay Study** — Hippocampal place cells that fire during maze navigation re-fire in the same sequence during subsequent sleep, strengthening the memory. Disrupting sleep replay impairs next-day performance. Evidence for sleep's causal role in memory consolidation
12. **Loftus “Lost in the Mall”** — Adults can be induced to “remember” being lost in a mall as children via suggestion. Demonstrates the reconstructive, fallible nature of episodic memory and forensic limits of eyewitness testimony
13. **Ebbinghaus’s Nonsense Syllables** — Self-experiments using meaningless syllables to chart the forgetting curve (exponential decay). Established experimental memory research and motivated spaced repetition
14. **Hemispatial Neglect** — Right parietal-lobe damage (typically stroke) → patient ignores the left half of space despite intact vision. Eats only from right side of plate, draws all clock numbers on right side. Proves attention is dissociable from sensation
15. **BCI Body Schema Extension** — With practice, BCI users’ motor maps reorganise to treat external prosthetics as part of their own body. Evidence for radical neural plasticity

Intelligence Theories — At a Glance

Theory		Proposer	Central Claim
Two-Factor ($g + s$)		Spearman (1904)	Single general factor g underlies all abilities, plus task-specific s
Primary Abilities	Mental	Thurstone (1938)	Seven independent primary abilities (verbal, spatial, number, memory, perceptual speed, reasoning, word fluency)
Multiple intelligences	Intelli-	Gardner (1983)	Eight+ independent intelligences (linguistic, logical-math, spatial, bodily-kinesthetic, musical, inter/intrapersonal, naturalistic)
Triarchic Theory		Sternberg	Analytical + creative + practical intelligence
Structure of Intellect		Guilford	Combinatorial: operations $\times$ contents $\times$ products
Fluid vs. tallised	Crys-	Cattell	Fluid (reasoning, declines with age) + crystallised (knowledge, stable with age)
Emotional intelligence	Intelli-	Goleman (1995)	Self-awareness, self-regulation, motivation, empathy, social skills

Memory Models — At a Glance

Model	Key Idea
Atkinson–Shiffrin (multi-store)	Sensory $\rightarrow$ STM (attention) $\rightarrow$ LTM (rehearsal); retrieval pulls LTM back to STM
Baddeley Working Memory	Central executive + phonological loop + visuospatial sketchpad + episodic buffer
Levels of Processing (Craik–Lockhart)	Shallow (surface) vs. deep (semantic); deep produces better retention
Ebbinghaus Forgetting Curve	Exponential/power-law decay; rapid initial loss, then stabilises; counteracted by spaced repetition
Hebbian / LTP	“Neurons that fire together, wire together”; synaptic strengthening is the cellular basis of memory
Systems Consolidation	Memories gradually transfer from hippocampus to distributed neocortex over days–years

Attention Models — At a Glance

Model / Concept		Key Idea
Broadbent's Filter	Early	Attention filter blocks unattended channels before semantic analysis
Deutsch–Norman Late Selection		All inputs processed semantically; selection is late
Treisman's Attenuation		Unattended channels attenuated (not blocked); salient content can break through
Feature Integration Theory		Preattentive features (parallel) + attentive binding (serial); solves the binding problem
Top-Down Bottom-Up	vs.	Goal-directed (prefrontal) vs. stimulus-driven (parietal/SC); operate simultaneously
Self-Terminating Exhaustive Search	vs.	Different RT signatures for target-present vs. absent trials
Cocktail Party Effect		Selective attention amid competing auditory streams; name detection illustrates attenuation, not total blocking
Hemispatial Neglect		Right parietal damage → left-space inattention; dissociates attention from sensation

Motor Control (Full Picture)

Concept	Role
Forward model	Predicts sensory consequences of a motor command
Inverse model / Control policy	Produces the motor command required for a desired state
State estimation	Infers current body/world state (parietal cortex)
Cost–reward evaluation	Selects best action (basal ganglia)
Open-loop / closed-loop	Without / with real-time feedback
Hierarchical control	Spinal (execution) + cortex/cerebellum (tactical) + pre-frontal (strategic)
Skill-learning stages	Cognitive → associative → automatic (Fitts & Posner); control shifts cortical → subcortical
Reinforcement learning	Model-based (plan with world model) vs. model-free (trial-and-error)
BCI — non-invasive	EEG-based; lower signal quality, no surgery
BCI — invasive	ECoG, Utah arrays, Neuralink; high signal quality; enables complex decoding

Core Concepts Checklist

March lectures:

- ☐ Introspection and its limitations
- ☐ Behaviourism: S–R associations, reflexes, limitations
- ☐ Cognitive Revolution: Tolman (cognitive maps), Miller (7 ± 2), Chomsky (UG)
- ☐ Schemas, scripts, and counterfactual reasoning
- ☐ Consciousness, qualia, unconscious processes
- ☐ CNS vs. PNS; sensory–motor loop
- ☐ Grey matter vs. white matter; cortex vs. spinal cord arrangement
- ☐ CSF: function and location
- ☐ Four lobes: frontal, parietal, occipital, temporal + insula
- ☐ Sulci, gyri, fissures (central sulcus, lateral fissure, etc.)

- ☐ Subcortical structures: amygdala, hippocampus, basal ganglia, thalamus, hypothalamus
- ☐ Cerebellum and brainstem functions
- ☐ Brain development milestones; frontal lobe maturation to age 21
- ☐ PET, MRI, EEG, MEG, fMRI, ECoG, DBS
- ☐ EEG frequency bands: delta, theta, alpha, beta, gamma
- ☐ fMRI BOLD signal; mental imagery experiments
- ☐ Brain–Computer Interfaces (BCIs); surface vs. implanted
- ☐ Neurons: dendrites, soma, axon; action potentials
- ☐ Synapses: electrochemical transmission; synaptic plasticity
- ☐ Hodgkin–Huxley: resting (-70 mV), threshold (-55 mV)
- ☐ Spatial scales (molecules to CNS) and temporal scales (ms to years)
- ☐ Brain vs. computer comparison (parallel vs. sequential, distributed vs. localised, etc.)
- ☐ Grandmother cell vs. distributed coding
- ☐ Nature vs. nurture: both contribute; synaptic refinement through experience
- ☐ Perception as interpretation; visual illusions
- ☐ Sensory transducers: conversion of physical signals to electrical signals
- ☐ Top-down vs. bottom-up processing (Kanizsa triangle, occluded text)
- ☐ Predictive processing / Bayesian brain hypothesis (prior, likelihood, posterior, prediction error)
- ☐ Memory taxonomy: sensory (iconic, echoic), short-term, long-term (declarative vs. non-declarative)
- ☐ Declarative memory: semantic (facts) vs. episodic (personal events)
- ☐ Non-declarative memory: procedural (motor skills, hard to acquire/erase) and priming (subliminal)
- ☐ Motor homunculus: non-linear topographic map (face and hands overrepresented)
- ☐ Emotions: survival instinct, sense of self, personalisation, empathy
- ☐ Open-loop vs. closed-loop motor control

- ☐ Reflexes: withdrawal reflex, crossed extensor reflex (Sherrington), knee-jerk (patellar) reflex
- ☐ Hierarchical motor control: execution (spinal), tactical (motor cortex/cerebellum), strategic (association cortex)
- ☐ Skill acquisition stages: conscious (high cognitive load) → automatic (low cognitive load)
- ☐ Motor control loop: state estimation, control policy, forward model, prediction error
- ☐ Cost–reward evaluation in motor planning; degrees of freedom problem
- ☐ Handedness: Edinburgh Handedness Inventory; left hemisphere dominant for language
- ☐ DBS for Parkinson’s: subthalamic nucleus stimulation; dramatic tremor reduction
- ☐ Goal-directed behaviour: sub-goal decomposition, monitoring, adjustment

April lectures — Motor Control Continued, Intelligence, Memory, Attention:

- ☐ Stationary vs. non-stationary systems
- ☐ Forward model (command → predicted sensory); inverse model (desired state → command)
- ☐ State estimation (parietal cortex); latent variables
- ☐ Cost–reward optimisation (basal ganglia); degrees of freedom problem
- ☐ Skill learning stages: cognitive → associative → automatic (Fitts & Posner)
- ☐ BCI applications: prosthetics, exoskeletons, FES, communication for ALS
- ☐ Invasive (ECoG, Utah array, Neuralink) vs. non-invasive (EEG) BCIs
- ☐ Body-schema extension via BCI (neural plasticity)
- ☐ Neural prosthetics: cochlear implant, retinal prosthetic
- ☐ Reinforcement learning: model-based (planning) vs. model-free (trial-and-error)
- ☐ Spearman’s g (general intelligence factor + s)
- ☐ Thurstone’s 7 primary mental abilities
- ☐ Gardner’s 8 multiple intelligences
- ☐ Sternberg’s triarchic theory: analytical + creative + practical
- ☐ Guilford’s structure of intellect (operations $\times$ contents $\times$ products)

- ☐ Cattell: fluid (declines with age) vs. crystallised (stable) intelligence
- ☐ IQ formula: $(\text{Mental Age} / \text{Chronological Age}) \times 100$; modern deviation scoring
- ☐ IQ tests: Binet–Simon, Stanford–Binet, WAIS, WISC
- ☐ Emotional intelligence (Goleman): 5 components (self-awareness, self-regulation, motivation, empathy, social skills)
- ☐ Developmental milestones: working memory by age 9, frontal lobe by 21
- ☐ Cognitive decline and aging; MMSE, MoCA
- ☐ Giftedness; early intervention
- ☐ Nature–nurture in IQ (twin and adoption studies; heritability 0.5–0.8)
- ☐ Three memory stages: encoding, storage, retrieval
- ☐ Atkinson–Shiffrin multi-store model
- ☐ Baddeley working memory: central executive, phonological loop, visuospatial sketchpad, episodic buffer
- ☐ Patient H.M. — anterograde amnesia; hippocampal role
- ☐ Anterograde vs. retrograde amnesia
- ☐ Synaptic consolidation vs. systems consolidation
- ☐ Factors influencing memory: attention, emotion, context, state, elaboration, repetition, sleep
- ☐ Context-dependent and state-dependent learning
- ☐ Levels of processing (Craik–Lockhart): shallow vs. deep
- ☐ Mnemonics: chunking, acronyms, method of loci, elaborative encoding, spaced repetition
- ☐ Familiarity (medial/perirhinal cortex) vs. recognition (hippocampus)
- ☐ Long-Term Potentiation (LTP), Long-Term Depression (LTD), Hebbian rule
- ☐ Sleep and memory: slow-wave (declarative) + REM (procedural/emotional); rat maze replay
- ☐ Context noise; multi-context learning benefits
- ☐ Memory distortion and false memories; Loftus studies
- ☐ Ebbinghaus forgetting curve; nonsense syllables
- ☐ Free recall vs. cued recall vs. recognition

- ☐ Modularity in memory storage
- ☐ Flashbulb memories; emotion–amygdala interaction
- ☐ Attention as limited resource; cognitive economy
- ☐ Self-terminating vs. exhaustive search (RT signatures)
- ☐ Selective attention; cocktail party effect
- ☐ Early (Broadbent) vs. late (Deutsch–Norman) selection; attenuation (Treisman)
- ☐ Top-down (goal-directed) vs. bottom-up (stimulus-driven) attention
- ☐ Feature Integration Theory (Treisman); binding problem; illusory conjunctions
- ☐ Hemispatial neglect (right parietal damage → left-space inattention)
- ☐ Brain areas for attention: parietal, prefrontal, visual cortex, pulvinar, ACC, superior colliculus
- ☐ Attention vs. intention distinction
- ☐ Attentional control (meta-cognition); ADHD, meditation training